|INCH-POUND|

MIL-M-28787/510
21 APRIL 1989

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

ARITHMETIC-LOGIC UNIT MODULE, KEY CODE BWE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic Arithmetic Logic Unit (ALU) module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3,Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 2.381 failures per million operating hours for class I or 5.999 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be BWE.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A	Isolation	II	X	10	INF	10	INF	10	INF	MOhm
B1-B6	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Vcc stress	IV	S	---------- Module shall pass all remaining tests ----------						
D1-D4	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	nA
D5	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	nA
D6-D8	Ibd	V	S	0.0	1.1	0.0	1.1	0.0	1.1	nA
D9	Ibd	V	S	0 0	1 0	0 0	1.0	0 0	1 0	nA
D10	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0 1	nA
D11	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	nA
D12	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	nA
D13	Ibd	V	S	0.0	0 1	0.0	0.1	0 0	0 1	nA
D14	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	nA
D15,D16	Ibd	V	S	0.0	1.0	0.0	1.0	0 0	1 0	nA
D17	Ibd	V	S	0.0	1.1	0.0	1.1	0.0	1.1	nA
D18-D21	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	nA
D22	Ibd	V	S	0.0	1.1	0.0	1.1	0.0	1.1	nA
D23-D25	Ibd	V	S	0.0	4.1	0.0	4.1	0.0	4.1	nA
D26-D28	Ibd	V	S	0.0	4.0	0.0	4.0	0.0	4.0	nA
D29-D36	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	nA
D37-D39	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	nA
D40	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	nA
D41	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	nA
D42	Ibd	V	S	0.0	4.0	0.0	4.0	0.0	4.0	nA
D43	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	nA
D44,D45	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	nA
D46-D48	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	nA
D49	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	nA
D50	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	nA
D51-D53	Ibd	V	S	0.0	1.1	0.0	1.1	0.0	1.1	nA
D54	Ibd	V	S	-0.02	3.02	-0.02	3.02	-0.02	3.02	nA
D55	Ibd	V	S	0.0	1.1	0.0	1.1	0.0	1.1	nA
D56	Ibd	V	S	0.0	1.2	0.0	1.2	0.0	1.2	nA
D57-D61	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	nA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
E1-E3	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E4	Ios	VI	S	-85.0	-30.0	-85.0	-30.0	-85.0	-30.0	mA
E5	Ios	VI	S	-130.0	-15.0	-130.0	-15.0	-130.0	-15.0	mA
E6	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E7	Ios	VI	S	-85.0	-30.0	-85.0	-30.0	-85.0	-30.0	mA
E8	Ios	VI	S	-130.0	-15.0	-130.0	-15.0	-130.0	-15.0	mA
E9	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E10	Ios	VI	S	-85.0	-30.0	-85.0	-30.0	-85.0	-30.0	mA
E11	Ios	VI	S	-130.0	-15.0	-130.0	-15.0	-130.0	-15.0	mA
E12	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E13	Ios	VI	S	-130.0	-15.0	-130.0	-15.0	-130.0	-15.0	mA
E14	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E15	Ios	VI	S	-85.0	-30.0	-85.0	-30.0	-85.0	-30.0	mA
E16	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E17	Ios	VI	S	-85.0	-30.0	-85.0	-30.0	-85.0	-30.0	mA
E18	Ios	VI	S	-130.0	-15.0	-130.0	-15.0	-130.0	-15.0	mA
E19	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E20	Ios	VI	S	-85.0	-30.0	-85.0	-30.0	-85.0	-30.0	mA
E21	Ios	VI	S	-130.0	-15.0	-130.0	-15.0	-130.0	-15.0	mA
E22	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E23	Ios	VI	S	-85.0	-30.0	-85.0	-30.0	-85.0	-30.0	mA
E24	Ios	VI	S	-130.0	-15.0	-130.0	-15.0	-130.0	-15.0	mA
E25	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E26	Ios	VI	S	-85.0	-30.0	-85.0	-30.0	-85.0	-30.0	mA
E27	Ios	VI	S	-130.0	-15.0	-130.0	-15.0	-130.0	-15.0	mA
E28	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E29	Ios	VI	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
E30,E31	Ios	VI	S	-85.0	-30.0	-85.0	-30.0	-85.0	-30.0	mA
E32	Ios	VI	S	-100.0	-20.0	-100.0	-20.0	-100.0	-20.0	mA
E33-E36	Ios	VI	S	-85.0	-30.0	-85.0	-30.0	-85.0	-30.0	mA
E37	Ios	VI	S	-120.0	-30.0	-120.0	-30.0	-120.0	-30.0	mA
E38	Ios	VI	S	-85.0	-30.0	-85.0	-30.0	-85.0	-30.0	mA
E39	Ios	VI	S	-130.0	-15.0	-130.0	-15.0	-130.0	-15.0	mA
E40	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA

- See footnotes at end of table.

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				Min	Max	Min	Max	Min	Max	
E41	Ios	VI	S	-85.0	-30.0	-85.0	-30.0	-85.0	-30.0	mA
E42	Ios	VI	S	-130.0	-15.0	-130.0	-15.0	-130.0	-15.0	mA
E43	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E44	Ios	VI	S	-85.0	-30.0	-85.0	-30.0	-85.0	-30.0	mA
E45	Ios	VI	S	-130.0	-15.0	-130.0	-15.0	-130.0	-15.0	mA
E46	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E47	Ios	VI	S	-85.0	-30.0	-85.0	-30.0	-85.0	-30.0	mA
E48	Ios	VI	S	-130.0	-15.0	-130.0	-15.0	-130.0	-15.0	mA
E49	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E50	Ios	VI	S	-85.0	-30.0	-85.0	-30.0	-85.0	-30.0	mA
E51	Ios	VI	S	-130.0	-15.0	-130.0	-15.0	-130.0	-15.0	mA
E52	Ios	VI	S	-240.0	-30.0	-240.0	-30.0	-240.0	-30.0	mA
E53	Ios	VI	S	-85.0	-30.0	-85.0	-30.0	-85.0	-30.0	mA
E54	Ios	VI	S	-130.0	-15.0	-130.0	-15.0	-130.0	-15.0	mA
E55	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E56	Ios	VI	S	-85.0	-30.0	-85.0	-30.0	-85.0	-30.0	mA
E57	Ios	VI	S	-130.0	-15.0	-130.0	-15.0	-130.0	-15.0	mA
E58	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E59	Ios	VI	S	-85.0	-30.0	-85.0	-30.0	-85.0	-30.0	mA
E60	Ios	VI	S	-130.0	-15.0	-130.0	-15.0	-130.0	-15.0	mA
E61	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
F1-F4	Iih	VII	X	-0.72	0.04	-0.72	0.04	-0.72	0.04	mA
F5	Iih	VII	X	-2.25	0.075	-2.25	0.075	-2.25	0.075	mA
F6-F8	Iih	VII	X	-0.85	0.045	-0.85	0.045	-0.85	0.045	mA
F9	Iih	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
F10	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F11	Iih	VII	X	-1.0	0.04	-1.0	0.04	-1.0	0.04	mA
F12	Iih	VII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	mA
F13	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F14	Iih	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F15,F16	Iih	VII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	mA
F17	Iih	VII	X	-0.85	0.045	-0.85	0.045	-0.85	0.045	mA
F18-F21	Iih	VII	X	-1.0	0.04	-1.0	0.04	-1.0	0.04	mA

- See footnotes at end of table.

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				Min	Max	Min	Max	Min	Max	
F22	Ith	VII	X	-0.85	0.045	-0.85	0.045	-0.85	0.045	mA
F23-F25	Ith	VII	X	-3.48	0.18	-3.48	0.18	-3.48	0.18	mA
F26-F28	Ith	VII	X	-1.44	0.08	-1.44	0.08	-1.44	0.08	mA
F29-F36	Ith	VII	X	-0.72	0.04	-0.72	0.04	-0.72	0.04	mA
F37-F39	Ith	VII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
F40	Ith	VII	X	-2.0	0.05	-2.0	0.05	-2 0	0.05	mA
F41	Ith	VII	X	-4.0	0.1	-4.0	0.1	-4.0	0.1	mA
F42	Ith	VII	X	-1.44	0.08	-1.44	0.08	-1.44	0.08	mA
F43	Ith	VII	X	-4.0	0.1	-4.0	0.1	-4.0	0.1	mA
F44	Ith	VII	X	-1.2	0.07	-1.2	0.07	-1 2	0.07	mA
F45	Ith	VII	X	-0.8	0.05	-0.8	0.05	-0.8	0.05	mA
F46,F47	Ith	VII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
F48	Ith	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F49	Ith	VII	X	-4.0	0.1	-4.0	0.1	-4.0	0.1	mA
F50	Ith	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
F51-F53	Ith	VII	X	-0.85	0.045	-0.85	0.045	-0.85	0.045	mA
F54	Ith	VII	X	-7.15	-0.462	-7.15	-0.462	-7.15	-0.462	mA
F55	Ith	VII	X	-2.6	0.07	-2.6	0.07	-2.6	0.07	mA
F56	Ith	VII	X	-3.2	0.09	-3.2	0.09	-3.2	0.09	mA
F57	Ith	VII	X	-2.0	0.05	-2.0	0,05	-2.0	0.05	mA
F58-F81	Ith	VII	X	-0.72	0.04	-0.72	0.04	-0.72	0.04	mA
G1-G4	Iil	VIII	X	-0.72	0.04	-0.72	0.04	-0.72	0.04	mA
G5	Iil	VIII	X	-2.25	0.075	-2.25	0.075	-2.25	0.075	mA
G6-G8	Iil	VIII	X	-0.85	0.045	-0.85	0.045	-0.85	0.045	mA
G9	Iil	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G10	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0 6	0.02	mA
G11	Iil	VIII	X	-1.0	0.04	-1.0	0.04	-1.0	0.04	mA
G12	Iil	VIII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	mA
G13	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G14	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1 2	0.04	mA
G15,G16	Iil	VIII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	mA
G17	Iil	VIII	X	-0.85	0.045	-0.85	0.045	-0.85	0.045	mA
G18-G21	Iil	VIII	X	-1.0	0.04	-1.0	0.04	-1.0	0.04	mA

- See footnotes at end of table.

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				Min	Max	Min	Max	Min	Max	
G22	Iil	VIII	X	-0.85	0.045	-0.85	0.045	-0.85	0.045	mA
G23-G25	Iil	VIII	X	-3.48	0.18	-3.48	0.18	-3.48	0.18	mA
G26-G28	Iil	VIII	X	-1.44	0.08	-1.44	0.08	-1.44	0.08	mA
G29-G36	Iil	VIII	X	-0.72	0.04	-0.72	0.04	-0.72	0.04	mA
G37-G39	Iil	VIII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
G40	Iil	VIII	X	-2.0	0.05	-2.0	0.05	-2 0	0.05	mA
G41	Iil	VIII	X	-4.0	0.1	-4.0	0.1	-4.0	0.1	mA
G42	Iil	VIII	X	-1.44	0.08	-1.44	0.08	-1.44	0.08	mA
G43	Iil	VIII	X	-4.0	0.1	-4.0	0.1	-4.0	0.1	mA
G44	Iil	VIII	X	-1.2	0.07	-1.2	0.07	-1 2	0.07	mA
G45	Iil	VIII	X	-0.8	0.05	-0.8	0.05	-0.8	0.05	mA
G46,G47	Iil	VIII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
G48	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G49	Iil	VIII	X	-4.0	0.1	-4.0	0.1	-4.0	0.1	mA
G50	Iil	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G51-G53	Iil	VIII	X	-0.85	0.045	-0.85	0.045	-0.85	0.045	mA
G54	Iil	VIII	X	-7.15	-0.462	-7.15	-0.462	-7.15	-0.462	mA
G55	Iil	VIII	X	-2.6	0.07	-2.6	0.07	-2.6	0.07	mA
G56	Iil	VIII	X	-3.2	0.08	-3.2	0.08	-3.2	0.08	mA
G57	Iil	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G58-G61	Iil	VIII	X	-0.72	0.04	-0.72	0.04	-0.72	0.04	mA
H1-H60	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J61	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
K1	Iozh	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
K2	Iozl	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
L1-L4	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L5-L11	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1 2	0 0	V
L12	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L13,L14	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L15,L16	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
L17-L25	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L26-L39	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L40,L41	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L42	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1 5	0.0	V
L43	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L44-L47	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L48-L56	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L57-L60	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1 5	0.0	V
M1	PSFC	XIII	S	3.8	7.0	3.8	7.0	3.8	7.0	uF
M2	PSFC	XIII	S	5.9	11.0	5.9	11.0	5.9	11.0	uF
N	Icc	XIV	X	850	2040	850	2040	850	2040	mA
P1,P2	Functional	XV	X	See table XV						
Q1,Q2	Dynamic	XVI	S	See table XVI						
Q3,Q4	Dynamic	XVI	Q	See table XVI						
R1-R4	Ci/o	XVIII	Q	2	50	2	50	2	50	pF
R5-R8	Ci/o	XVIII	Q	4	100	4	100	4	100	pF
R9	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R10	Ci/o	XVIII	Q	2	50	2	50	2	50	pF
R11	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R12,R13	Ci/o	XVIII	Q	2	50	2	50	2	50	pF
R14	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R15	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R16	Ci/o	XVIII	Q	2	50	2	50	2	50	pF
R17	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R18	Ci/o	XVIII	Q	4	100	4	100	4	100	pF
R19-R22	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R23	Ci/o	XVIII	Q	6	200	6	200	6	200	pF
R24	Ci/o	XVIII	Q	10	150	10	150	10	150	pF
R25	Ci/o	XVIII	Q	6	175	6	175	6	175	pF
R26	Ci/o	XVIII	Q	4	150	4	150	4	150	pF

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
R27	C1/o	XVIII	Q	8	150	8	150	8	150	pF
R28	C1/o	XVIII	Q	4	150	4	150	4	150	pF
R29-R40	C1/o	XVIII	Q	2	50	2	50	2	50	pF
R41	C1/o	XVIII	Q	4	100	4	100	4	100	pF
R42	C1/o	XVIII	Q	6	150	6	150	6	150	pF
R43	C1/o	XVIII	Q	2	100	2	100	2	100	pF
R44-R46	C1/o	XVIII	Q	2	50	2	50	2	50	pF
R47,R48	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R49	C1/o	XVIII	Q	4	100	4	100	4	100	pF
R50	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R51,R52	C1/o	XVIII	Q	4	75	4	75	4	75	pF
R53	C1/o	XVIII	Q	4	100	4	100	4	100	pF
R54	C1/o	XVIII	Q	8	100	8	100	8	100	pF
R55	C1/o	XVIII	Q	4	75	4	75	4	75	pF
R56	C1/o	XVIII	Q	6	100	6	100	6	100	pF
R57	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R58-R61	C1/o	XVIII	Q	2	50	2	50	2	50	pF
R62	C1/o	XVIII	Q	10	150	10	150	10	150	pF
R63	C1/o	XVIII	Q	8	150	8	150	8	150	pF
R64,R65	C1/o	XVIII	Q	8	125	8	125	8	125	pF
R66-R77	C1/o	XVIII	Q	8	150	8	150	8	150	pF
T1,T2	PSTR	XIX	Q	0	400	0	400	0	400	mV

- See footnotes at end of table.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/510-1 for class I, and M28787/510-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

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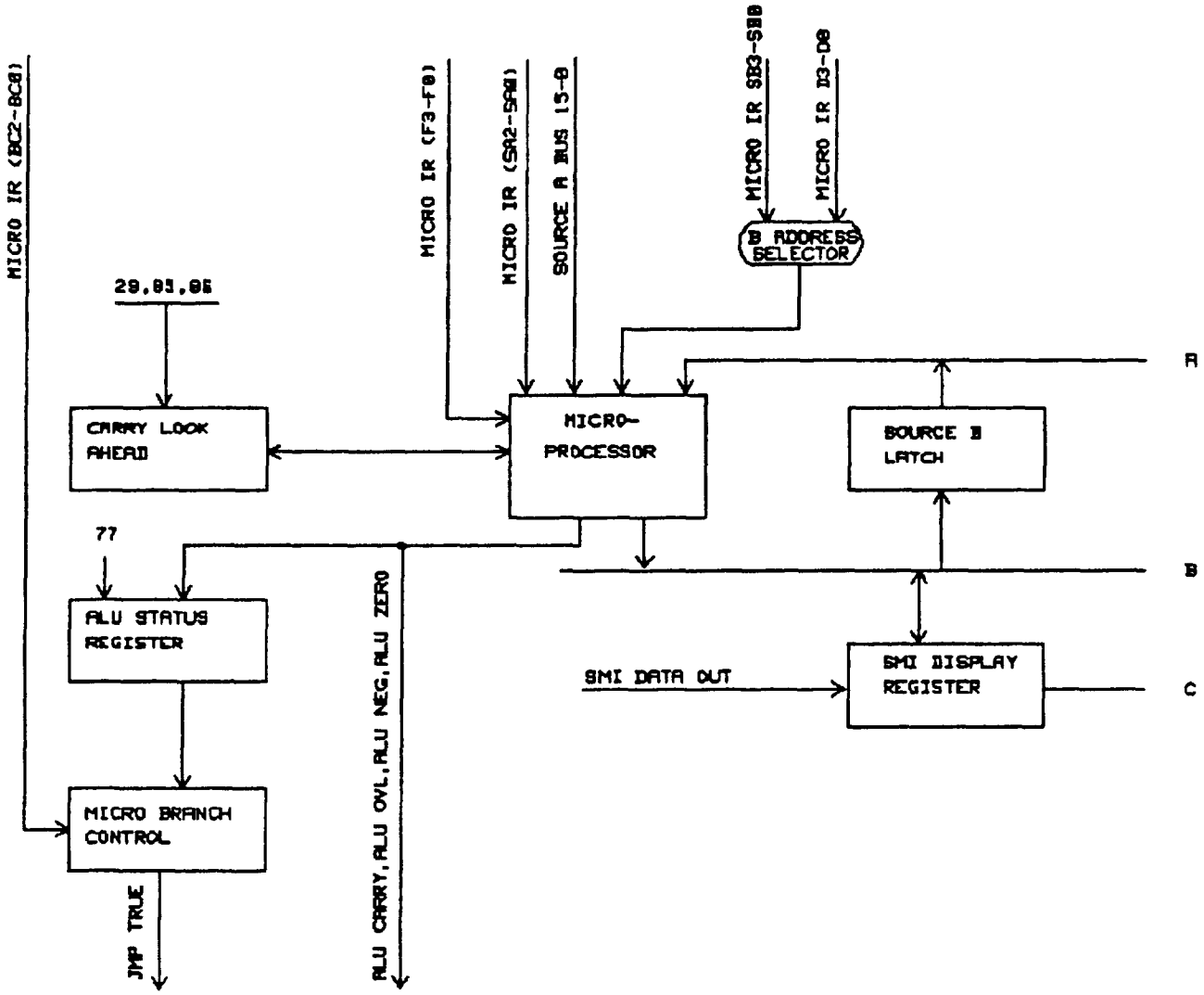


FIGURE 1. BWE functional block diagram.

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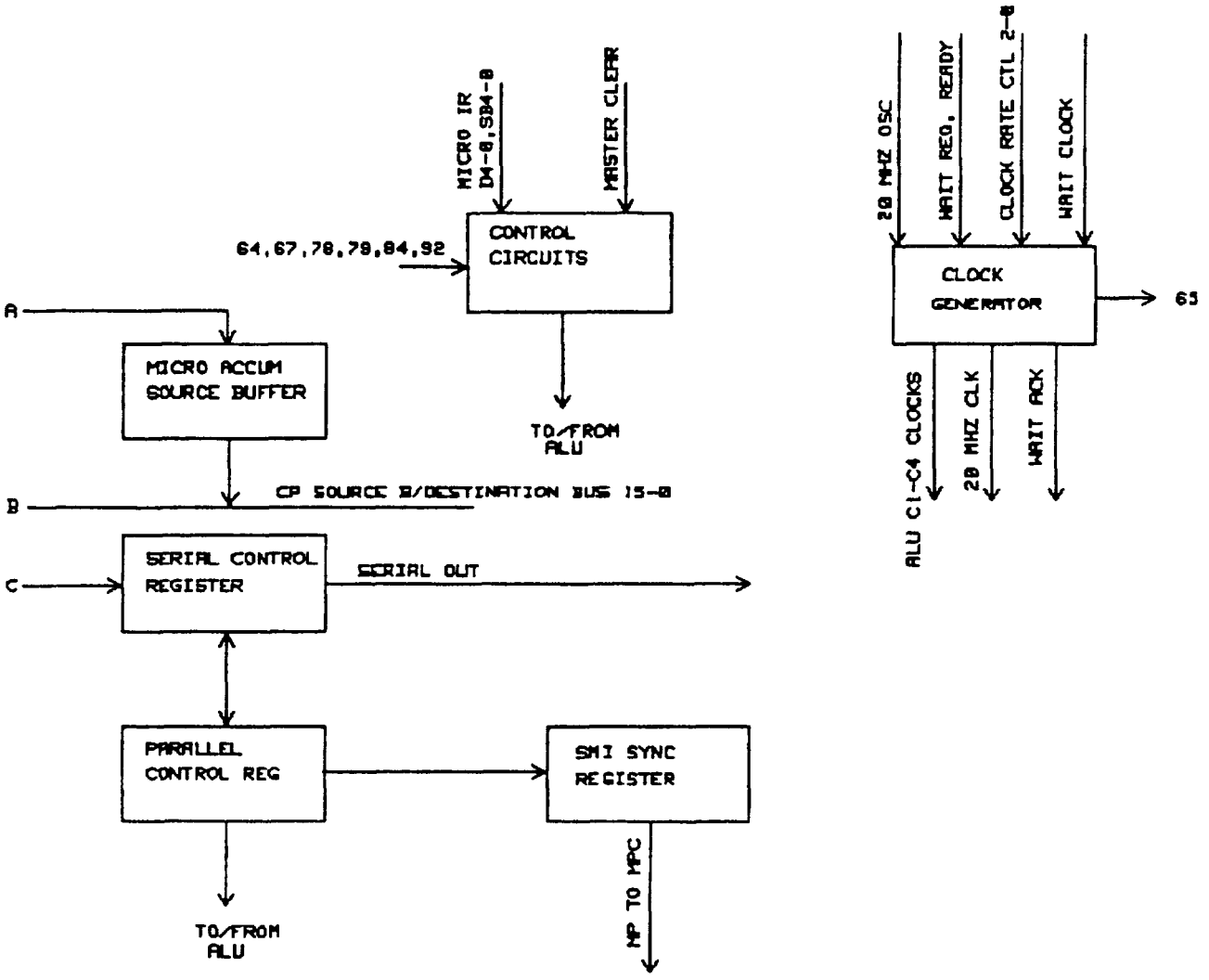


FIGURE 1. BWE functional block diagram - Continued.

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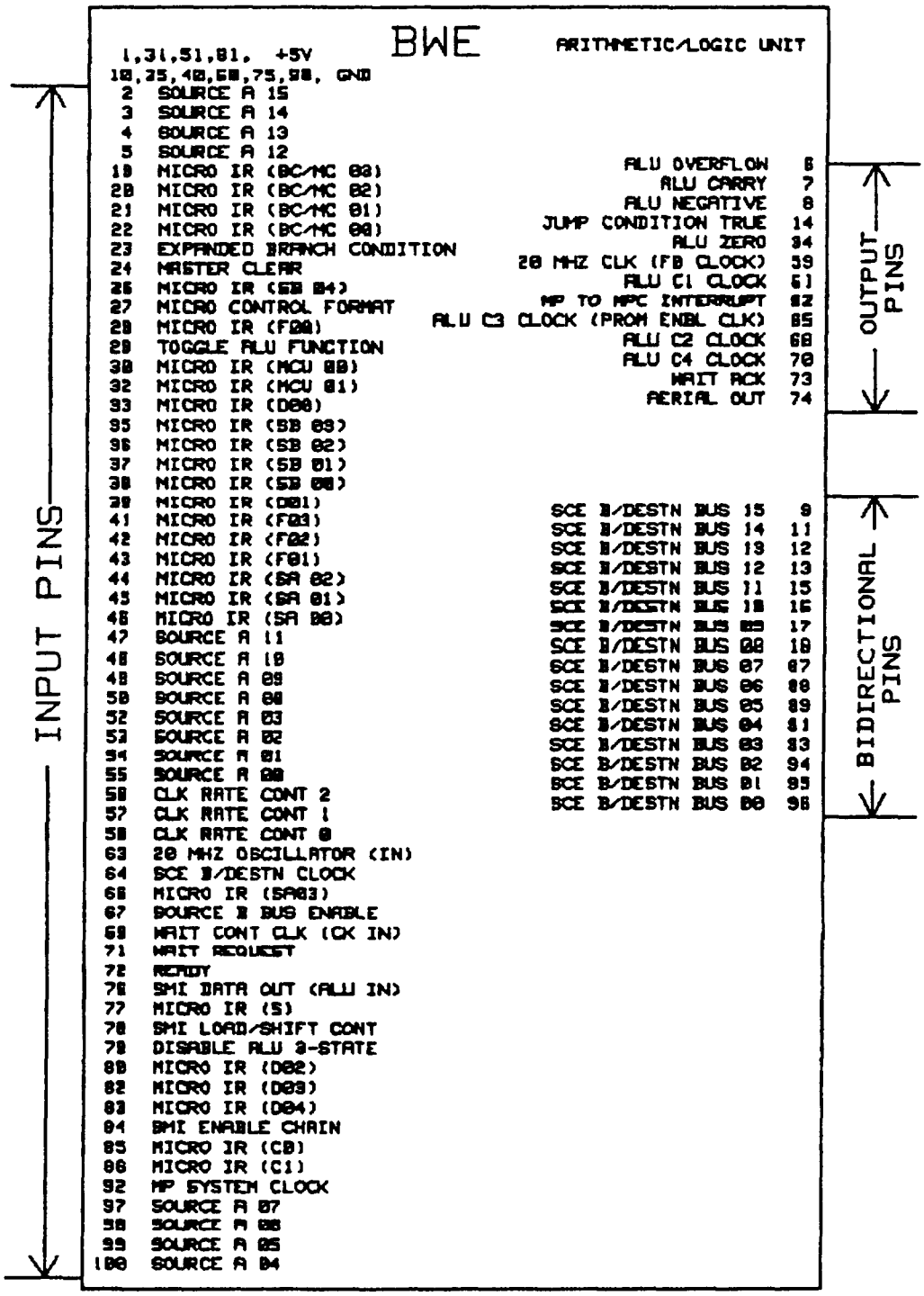


FIGURE 2. BWE pin assignment.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, and L are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or L are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, and L where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

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4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

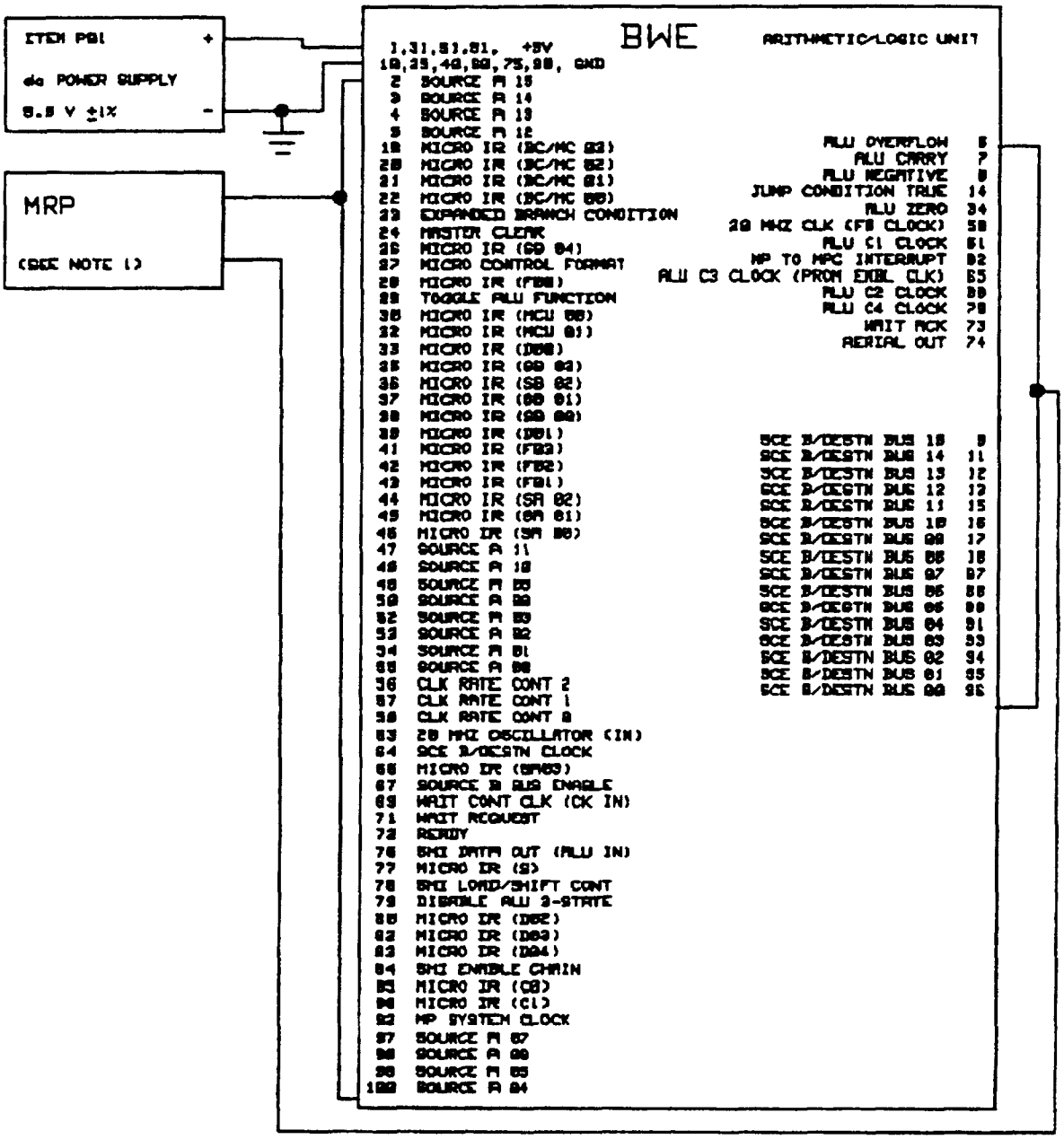
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. BWE operating and component temperature test circuit.

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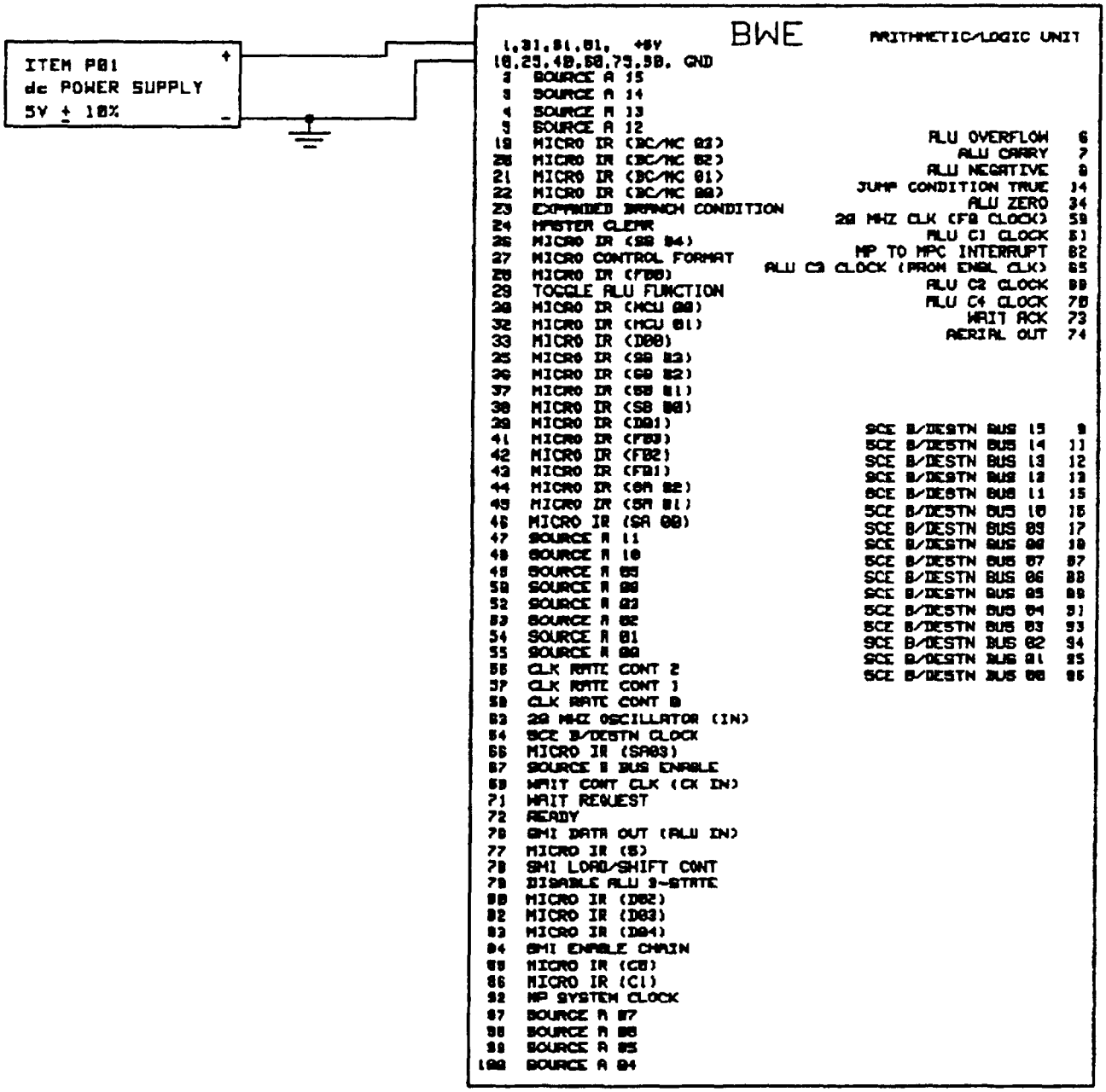
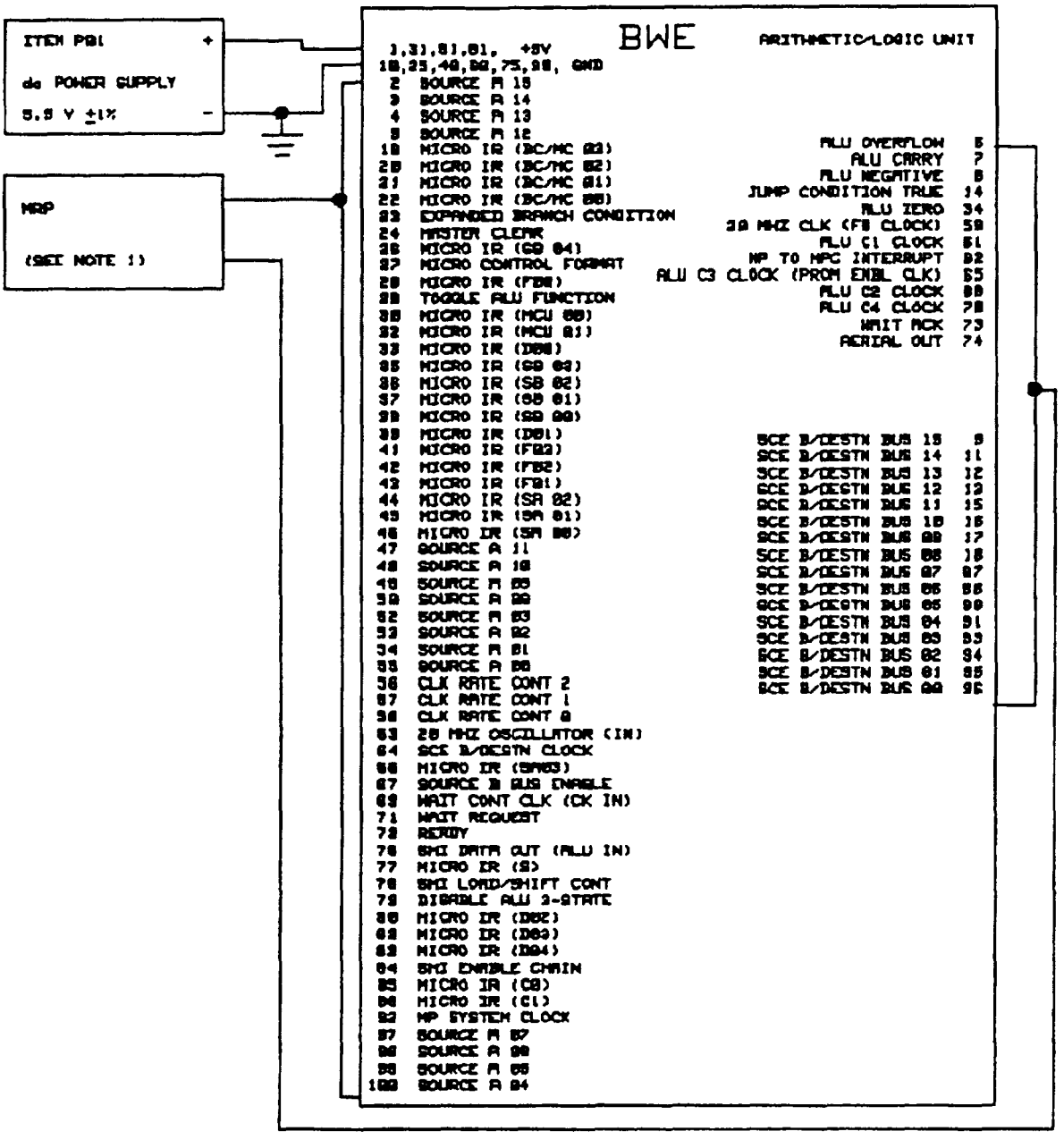


FIGURE 4. BWE vibration test circuit.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. BWE life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) Iozh = output high level leakage current.
(m) Iozl = output low level leakage current.
(n) NA = not applicable.
(o) PSFC = power supply filter capacitance.
(p) PSTA = power supply transient amplitude.
(q) Vic = input clamp diode voltage.
(r) Vih = input high level voltage.
(s) Vil = input low level voltage.
(t) Vin = input forcing voltage.
(u) Voh = output high level voltage.
(v) Vol = output low level voltage.
(w) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 11.22 W (maximum), 6.74 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
DESTN BUS 15	NA	0.4	2.4	NA	NA	NA	150
DESTN BUS 14	NA	0.4	2.4	NA	NA	NA	150
DESTN BUS 13	NA	0.4	2.4	NA	NA	NA	125
DESTN BUS 12	NA	0.4	2.4	NA	NA	NA	125
DESTN BUS 11	NA	0.4	2.4	NA	NA	NA	150
DESTN BUS 10	NA	0.4	2.4	NA	NA	NA	150
DESTN BUS 9	NA	0.4	2.4	NA	NA	NA	150
DESTN BUS 8	NA	0.4	2.4	NA	NA	NA	150
DESTN BUS 7	NA	0.4	2.4	NA	NA	NA	150
DESTN BUS 6	NA	0.4	2.4	NA	NA	NA	150
DESTN BUS 5	NA	0.4	2.4	NA	NA	NA	150
DESTN BUS 4	NA	0.4	2.4	NA	NA	NA	150
DESTN BUS 3	NA	0.4	2.4	NA	NA	NA	150
DESTN BUS 2	NA	0.4	2.4	NA	NA	NA	150

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
DESTN BUS 1	NA	0.4	2.4	NA	NA	NA	150
DESTN BUS 0	NA	0.4	2.4	NA	NA	NA	150
SOURCE A 15	-1.5	0.4	2.4	-0.72	0.04	1.0	50
SOURCE A 14	-1.5	0.4	2.4	-0.72	0.04	1.0	50
SOURCE A 13	-1.5	0.4	2.4	-0.72	0.04	1.0	50
SOURCE A 12	-1.5	0.4	2.4	-0.72	0.04	1.0	50
BC/MC 03	-1.2	0.4	2.4	-2.25	0.075	2.0	100
BC/MC 02	-1.2	0.4	2.4	-0.85	0.045	1.1	100
BC/MC 01	-1.2	0.4	2.4	-0.85	0.045	1.1	100
BC/MC 00	-1.2	0.4	2.4	-0.85	0.045	1.1	100
EX BRAN COND	-1.2	0.4	2.4	-2.0	0.05	1.0	75
MASTER CLEAR	-1.2	0.4	2.4	-0.6	0.02	0.1	50
(SB04)	-1.2	0.4	2.4	-1.0	0.04	0.2	75
MC CL FORMAT	-1.5	0.4	2.4	-0.25	0.025	1.0	50

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
MC IR (F00)	-1.2	0.4	2.4	-0.6	0.02	0.1	50
TOGGLE ALU	-1.2	0.4	2.4	-1.2	0.04	0.2	75
MCU (00)	-1.5	0.4	2.4	-0.25	0.025	1.0	75
MCU (01)	-1.5	0.4	2.4	-0.25	0.025	1.0	50
(D00)	-1.2	0.4	2.4	-0.85	0.045	1.1	75
(SB03)	-1.2	0.4	2.4	-1.0	0.04	0.2	100
(SB02)	-1.2	0.4	2.4	-1.0	0.04	0.2	75
(SB01)	-1.2	0.4	2.4	-1.0	0.04	0.2	75
(SB00)	-1.2	0.4	2.4	-1.0	0.04	0.2	75
(D01)	-1.2	0.4	2.4	-0.85	0.045	1.1	75
(F03)	-1.2	0.4	2.4	-3.48	0.18	4.1	200
(F02)	-1.2	0.4	2.4	-3.48	0.18	4.1	150
(F01)	-1.2	0.4	2.4	-3.48	0.18	4.1	175
(SA02)	-1.5	0.4	2.4	-1.44	0.08	4.0	150

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
(SA01)	-1.5	0.4	2.4	-1.44	0.08	4.0	150
(SA00)	-1.5	0.4	2.4	-1.44	0.08	4.0	150
SOURCE A 11	-1.5	0.4	2.4	-0.72	0.04	1.0	50
SOURCE A 10	-1.5	0.4	2.4	-0.72	0.04	1.0	50
SOURCE A 9	-1.5	0.4	2.4	-0.72	0.04	1.0	50
SOURCE A 8	-1.5	0.4	2.4	-0.72	0.04	1.0	50
SOURCE A 3	-1.5	0.4	2.4	-0.72	0.04	1.0	50
SOURCE A 2	-1.5	0.4	2.4	-0.72	0.04	1.0	50
SOURCE A 1	-1.5	0.4	2.4	-0.72	0.04	1.0	50
SOURCE A 0	-1.5	0.4	2.4	-0.72	0.04	1.0	50
CLK RATE 2	-1.5	0.4	2.4	-0.4	0.02	0.1	50
CLK RATE 1	-1.5	0.4	2.4	-0.4	0.02	0.1	50
CLK RATE 0	-1.5	0.4	2.4	-0.4	0.02	0.1	50
20 MHZ OSC	-1.2	0.4	2.4	-2.0	0.05	1.0	50

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
DESTN CK	-1.2	0.4	2.4	-4.0	0.1	2.0	100
(SA03)	-1.5	0.4	2.4	-1.44	0.08	4.0	150
B BUS ENABLE	-1.2	0.4	2.4	-4.0	0.1	2.0	100
WAIT CLOCK	-1.5	0.4	2.4	-1.2	0.07	1.0	50
WAIT REQUEST	-1.5	0.4	2.4	-0.8	0.05	1.0	50
READY	-1.5	0.4	2.4	-0.4	0.02	0.1	50
SMI DATA OUT	-1.5	0.4	2.4	-0.4	0.02	0.1	75
MICRO IR (S)	-1.2	0.4	2.4	-0.6	0.02	0.1	75
SMI LD/SF CT	-1.2	0.4	2.4	-4.0	0.1	2.0	100
DISABLE ALU	-1.2	0.4	2.4	-2.0	0.05	1.0	75
(D02)	-1.2	0.4	2.4	-0.85	0.045	1.1	75
(D03)	-1.2	0.4	2.4	-0.85	0.045	1.1	75
(D04)	-1.2	0.4	2.4	-0.85	0.045	1.1	100
ENABLE CHAIN	NA	0.4	2.4	-7.15	-0.462	3.02	100

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
(C0)	-1.2	0.4	2.4	-2.6	0.07	1.1	75
(C1)	-1.2	0.4	2.4	-3.2	0.09	1.2	100
SYS CLOCK	-1.2	0.4	2.4	-2.0	0.05	1.0	75
SOURCE A 7	-1.5	0.4	2.4	-0.72	0.04	1.0	50
SOURCE A 6	-1.5	0.4	2.4	-0.72	0.04	1.0	50
SOURCE A 5	-1.5	0.4	2.4	-0.72	0.04	1.0	50
SOURCE A 4	-1.5	0.4	2.4	-0.72	0.04	1.0	50

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
DESTN BUS 15	2.4	0.5	-0.16	1.6	NA	NA	-30.0	150
DESTN BUS 15	2.4	0.5	-0.16	1.6	NA	NA	-15.0	150
DESTN BUS 14	2.4	0.5	-0.16	1.6	NA	NA	-30.0	150
DESTN BUS 14	2.4	0.5	-0.16	1.6	NA	NA	-15.0	150
DESTN BUS 13	2.4	0.5	-0.16	1.6	NA	NA	-30.0	125
DESTN BUS 13	2.4	0.5	-0.16	1.6	NA	NA	-15.0	125
DESTN BUS 12	2.4	0.5	-0.16	1.6	NA	NA	-30.0	125
DESTN BUS 12	2.4	0.5	-0.16	1.6	NA	NA	-15.0	125
DESTN BUS 11	2.4	0.5	-0.16	1.6	NA	NA	-30.0	150
DESTN BUS 11	2.4	0.5	-0.16	1.6	NA	NA	-15.0	150
DESTN BUS 10	2.4	0.5	-0.16	1.6	NA	NA	-30.0	150
DESTN BUS 10	2.4	0.5	-0.16	1.6	NA	NA	-15.0	150
DESTN BUS 9	2.4	0.5	-0.16	1.6	NA	NA	-30.0	150
DESTN BUS 9	2.4	0.5	-0.16	1.6	NA	NA	-15.0	150

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
DESTN BUS 8	2.4	0.5	-0.16	1.6	NA	NA	-30.0	150
DESTN BUS 8	2.4	0.5	-0.16	1.6	NA	NA	-15.0	150
DESTN BUS 7	2.4	0.5	-0.16	1.6	NA	NA	-30.0	150
DESTN BUS 7	2.4	0.5	-0.16	1.6	NA	NA	-15.0	150
DESTN BUS 6	2.4	0.5	-0.16	1.6	NA	NA	-30.0	150
DESTN BUS 6	2.4	0.5	-0.16	1.6	NA	NA	-15.0	150
DESTN BUS 5	2.4	0.5	-0.16	1.6	NA	NA	-30.0	150
DESTN BUS 5	2.4	0.5	-0.16	1.6	NA	NA	-15.0	150
DESTN BUS 4	2.4	0.5	-0.16	1.6	NA	NA	-30.0	150
DESTN BUS 4	2.4	0.5	-0.16	1.6	NA	NA	-15.0	150
DESTN BUS 3	2.4	0.5	-0.16	1.6	NA	NA	-30.0	150
DESTN BUS 3	2.4	0.5	-0.16	1.6	NA	NA	-15.0	150
DESTN BUS 2	2.4	0.5	-0.16	1.6	NA	NA	-30.0	150
DESTN BUS 2	2.4	0.5	-0.16	1.6	NA	NA	-15.0	150

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
DESTN BUS 1	2.4	0.5	-0.16	1.6	NA	NA	-30.0	150
DESTN BUS 1	2.4	0.5	-0.16	1.6	NA	NA	-15.0	150
DESTN BUS 0	2.4	0.5	-0.16	1.6	NA	NA	-30.0	150
DESTN BUS 0	2.4	0.5	-0.16	1.6	NA	NA	-15.0	150
ALU OVL	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
ALU CARRY	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
ALU NEG	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
JMP TRUE	2.4	0.5	-0.4	6.0	0.05	0.05	-60.0	NA
ALU ZERO	2.4	0.5	-0.4	6.0	NA	NA	-40.0	NA
20 MHZ CLOCK	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
ALU C1 CLOCK	2.4	0.5	-0.4	1.6	NA	NA	-30.0	NA
MP TO MPC	2.4	0.5	-0.16	1.6	NA	NA	-20.0	NA
ALU C3 CLOCK	2.4	0.5	-0.4	1.6	NA	NA	-30.0	NA
ALU C2 CLOCK	2.4	0.5	-0.4	1.6	NA	NA	-30.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
ALU C4 CLOCK	2.4	0.5	-0.4	1.6	NA	NA	-30.0	NA
WAIT ACK	2.4	0.5	-0.4	1.6	NA	NA	-30.0	NA
SERIAL OUT	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	DESTN CK to DESTN BUS 15	-	-	-	85	85	85	nS	12
1	tpd	DESTN CK to DESTN BUS 14	-	-	-	85	85	85	nS	12
1	tpd	DESTN CK to DESTN BUS 13	-	-	-	85	85	85	nS	12
1	tpd	DESTN CK to DESTN BUS 12	-	-	-	85	85	85	nS	12
1	tpd	DESTN CK to DESTN BUS 11	-	-	-	85	85	85	nS	12
1	tpd	DESTN CK to DESTN BUS 10	-	-	-	85	85	85	nS	12
1	tpd	DESTN CK to DESTN BUS 9	-	-	-	85	85	85	nS	12
1	tpd	DESTN CK to DESTN BUS 8	-	-	-	85	85	85	nS	12
1	tpd	DESTN CK to DESTN BUS 7	-	-	-	85	85	85	nS	12
1	tpd	DESTN CK to DESTN BUS 6	-	-	-	85	85	85	nS	12
1	tpd	DESTN CK to DESTN BUS 5	-	-	-	85	85	85	nS	12
1	tpd	DESTN CK to DESTN BUS 4	-	-	-	85	85	85	nS	12
1	tpd	DESTN CK to DESTN BUS 3	-	-	-	85	85	85	nS	12
1	tpd	DESTN CK to DESTN BUS 2	-	-	-	85	85	85	nS	12
1	tpd	DESTN CK to DESTN BUS 1	-	-	-	85	85	85	nS	12
1	tpd	DESTN CK to DESTN BUS 0	-	-	-	85	85	85	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tpd	(SB04) to DESTN BUS 15	-	-	-	165	165	165	nS	12
2	tpd	(SB04) to DESTN BUS 14	-	-	-	165	165	165	nS	12
2	tpd	(SB04) to DESTN BUS 13	-	-	-	165	165	165	nS	12
2	tpd	(SB04) to DESTN BUS 12	-	-	-	165	165	165	nS	12
2	tpd	(SB04) to DESTN BUS 11	-	-	-	165	165	165	nS	12
2	tpd	(SB04) to DESTN BUS 10	-	-	-	165	165	165	nS	12
2	tpd	(SB04) to DESTN BUS 9	-	-	-	165	165	165	nS	12
2	tpd	(SB04) to DESTN BUS 8	-	-	-	165	165	165	nS	12
2	tpd	(SB04) to DESTN BUS 7	-	-	-	165	165	165	nS	12
2	tpd	(SB04) to DESTN BUS 6	-	-	-	165	165	165	nS	12
2	tpd	(SB04) to DESTN BUS 5	-	-	-	165	165	165	nS	12
2	tpd	(SB04) to DESTN BUS 4	-	-	-	165	165	165	nS	12
		to DESTN BUS 3								

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tpd	(SB04) to DESTN BUS 2	-	-	-	165	165	165	nS	12
2	tpd	(SB04) to DESTN BUS 1	-	-	-	165	165	165	nS	12
2	tpd	(SB04) to DESTN BUS 0	-	-	-	165	165	165	nS	12
2	tpd	MC IR (F00) to DESTN BUS 15	-	-	-	165	165	165	nS	12
2	tpd	MC IR (F00) to DESTN BUS 14	-	-	-	165	165	165	nS	12
2	tpd	MC IR (F00) to DESTN BUS 13	-	-	-	165	165	165	nS	12
2	tpd	MC IR (F00) to DESTN BUS 12	-	-	-	165	165	165	nS	12
2	tpd	MC IR (F00) to DESTN BUS 11	-	-	-	165	165	165	nS	12
2	tpd	MC IR (F00) to DESTN BUS 10	-	-	-	165	165	165	nS	12
2	tpd	MC IR (F00) to DESTN BUS 9	-	-	-	165	165	165	nS	12
2	tpd	MC IR (F00) to DESTN BUS 8	-	-	-	165	165	165	nS	12
2	tpd	MC IR (F00) to DESTN BUS 7	-	-	-	165	165	165	nS	12
2	tpd	MC IR (F00) to DESTN BUS 6	-	-	-	165	165	165	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tpd	MC IR (F00) to DESTN BUS 5	-	-	-	165	165	165	nS	12
2	tpd	MC IR (F00) to DESTN BUS 4	-	-	-	165	165	165	nS	12
2	tpd	MC IR (F00) to DESTN BUS 3	-	-	-	165	165	165	nS	12
2	tpd	MC IR (F00) to DESTN BUS 2	-	-	-	165	165	165	nS	12
2	tpd	MC IR (F00) to DESTN BUS 1	-	-	-	165	165	165	nS	12
2	tpd	MC IR (F00) to DESTN BUS 0	-	-	-	165	165	165	nS	12
2	tpd	(SB03) to DESTN BUS 15	-	-	-	165	165	165	nS	12
2	tpd	(SB03) to DESTN BUS 14	-	-	-	165	165	165	nS	12
2	tpd	(SB03) to DESTN BUS 13	-	-	-	165	165	165	nS	12
2	tpd	(SB03) to DESTN BUS 12	-	-	-	165	165	165	nS	12
2	tpd	(SB03) to DESTN BUS 11	-	-	-	165	165	165	nS	12
2	tpd	(SB03) to DESTN BUS 10	-	-	-	165	165	165	nS	12
2	tpd	(SB03) to DESTN BUS 9	-	-	-	165	165	165	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tpd	(SB03) to DESTN BUS 8	-	-	-	165	165	165	nS	12
2	tpd	(SB03) to DESTN BUS 7	-	-	-	165	165	165	nS	12
2	tpd	(SB03) to DESTN BUS 6	-	-	-	165	165	165	nS	12
2	tpd	(SB03) to DESTN BUS 5	-	-	-	165	165	165	nS	12
2	tpd	(SB03) to DESTN BUS 4	-	-	-	165	165	165	nS	12
2	tpd	(SB03) to DESTN BUS 3	-	-	-	165	165	165	nS	12
2	tpd	(SB03) to DESTN BUS 2	-	-	-	165	165	165	nS	12
2	tpd	(SB03) to DESTN BUS 1	-	-	-	165	165	165	nS	12
2	tpd	(SB03) to DESTN BUS 0	-	-	-	165	165	165	nS	12
2	tpd	(SB02) to DESTN BUS 15	-	-	-	165	165	165	nS	12
2	tpd	(SB02) to DESTN BUS 14	-	-	-	165	165	165	nS	12
2	tpd	(SB02) to DESTN BUS 13	-	-	-	165	165	165	nS	12
2	tpd	(SB02) to DESTN BUS 12	-	-	-	165	165	165	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tpd	(SB02) to DESTN BUS 11	-	-	-	165	165	165	nS	12
2	tpd	(SB02) to DESTN BUS 10	-	-	-	165	165	165	nS	12
2	tpd	(SB02) to DESTN BUS 9	-	-	-	165	165	165	nS	12
2	tpd	(SB02) to DESTN BUS 8	-	-	-	165	165	165	nS	12
2	tpd	(SB02) to DESTN BUS 7	-	-	-	165	165	165	nS	12
2	tpd	(SB02) to DESTN BUS 6	-	-	-	165	165	165	nS	12
2	tpd	(SB02) to DESTN BUS 5	-	-	-	165	165	165	nS	12
2	tpd	(SB02) to DESTN BUS 4	-	-	-	165	165	165	nS	12
2	tpd	(SB02) to DESTN BUS 3	-	-	-	165	165	165	nS	12
2	tpd	(SB02) to DESTN BUS 2	-	-	-	165	165	165	nS	12
2	tpd	(SB02) to DESTN BUS 1	-	-	-	165	165	165	nS	12
2	tpd	(SB01) to DESTN BUS 15	-	-	-	165	165	165	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tpd	(SB01) to DESTN BUS 14	-	-	-	165	165	165	nS	12
2	tpd	(SB01) to DESTN BUS 13	-	-	-	165	165	165	nS	12
2	tpd	(SB01) to DESTN BUS 12	-	-	-	165	165	165	nS	12
2	tpd	(SB01) to DESTN BUS 11	-	-	-	165	165	165	nS	12
2	tpd	(SB01) to DESTN BUS 10	-	-	-	165	165	165	nS	12
2	tpd	(SB01) to DESTN BUS 9	-	-	-	165	165	165	nS	12
2	tpd	(SB01) to DESTN BUS 8	-	-	-	165	165	165	nS	12
2	tpd	(SB01) to DESTN BUS 7	-	-	-	165	165	165	nS	12
2	tpd	(SB01) to DESTN BUS 6	-	-	-	165	165	165	nS	12
2	tpd	(SB01) to DESTN BUS 5	-	-	-	165	165	165	nS	12
2	tpd	(SB01) to DESTN BUS 4	-	-	-	165	165	165	nS	12
2	tpd	(SB01) to DESTN BUS 3	-	-	-	165	165	165	nS	12
2	tpd	(SB01) to DESTN BUS 2	-	-	-	165	165	165	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tpd	(SB01) to DESTN BUS 1	-	-	-	165	165	165	nS	12
2	tpd	(SB01) to DESTN BUS 0	-	-	-	165	165	165	nS	12
2	tpd	(SB00) to DESTN BUS 15	-	-	-	165	165	165	nS	12
2	tpd	(SB00) to DESTN BUS 14	-	-	-	165	165	165	nS	12
2	tpd	(SB00) to DESTN BUS 13	-	-	-	165	165	165	nS	12
2	tpd	(SB00) to DESTN BUS 12	-	-	-	165	165	165	nS	12
2	tpd	(SB00) to DESTN BUS 11	-	-	-	165	165	165	nS	12
2	tpd	(SB00) to DESTN BUS 10	-	-	-	165	165	165	nS	12
2	tpd	(SB00) to DESTN BUS 9	-	-	-	165	165	165	nS	12
2	tpd	(SB00) to DESTN BUS 8	-	-	-	165	165	165	nS	12
2	tpd	(SB00) to DESTN BUS 7	-	-	-	165	165	165	nS	12
2	tpd	(SB00) to DESTN BUS 6	-	-	-	165	165	165	nS	12
		to DESTN BUS 5								

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tpd	(SB00) to DESTN BUS 4	-	-	-	165	165	165	nS	12
2	tpd	(SB00) to DESTN BUS 3	-	-	-	165	165	165	nS	12
2	tpd	(SB00) to DESTN BUS 2	-	-	-	165	165	165	nS	12
2	tpd	(SB00) to DESTN BUS 1	-	-	-	165	165	165	nS	12
2	tpd	(SB00) to DESTN BUS 0	-	-	-	165	165	165	nS	12
2	tpd	(F03) to DESTN BUS 15	-	-	-	165	165	165	nS	12
2	tpd	(F03) to DESTN BUS 14	-	-	-	165	165	165	nS	12
2	tpd	(F03) to DESTN BUS 13	-	-	-	165	165	165	nS	12
2	tpd	(F03) to DESTN BUS 12	-	-	-	165	165	165	nS	12
2	tpd	(F03) to DESTN BUS 11	-	-	-	165	165	165	nS	12
2	tpd	(F03) to DESTN BUS 10	-	-	-	165	165	165	nS	12
2	tpd	(F03) to DESTN BUS 9	-	-	-	165	165	165	nS	12
2	tpd	(F03) to DESTN BUS 8	-	-	-	165	165	165	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tpd	(F03) to DESTN BUS 7	-	-	-	165	165	165	nS	12
2	tpd	(F03) to DESTN BUS 6	-	-	-	165	165	165	nS	12
2	tpd	(F03) to DESTN BUS 5	-	-	-	165	165	165	nS	12
2	tpd	(F03) to DESTN BUS 4	-	-	-	165	165	165	nS	12
2	tpd	(F03) to DESTN BUS 3	-	-	-	165	165	165	nS	12
2	tpd	(F03) to DESTN BUS 2	-	-	-	165	165	165	nS	12
2	tpd	(F03) to DESTN BUS 1	-	-	-	165	165	165	nS	12
2	tpd	(F03) to DESTN BUS 0	-	-	-	165	165	165	nS	12
2	tpd	(F02) to DESTN BUS 15	-	-	-	165	165	165	nS	12
2	tpd	(F02) to DESTN BUS 14	-	-	-	165	165	165	nS	12
2	tpd	(F02) to DESTN BUS 13	-	-	-	165	165	165	nS	12
2	tpd	(F02) to DESTN BUS 12	-	-	-	165	165	165	nS	12
2	tpd	(F02) to DESTN BUS 11	-	-	-	165	165	165	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tpd	(F02) to DESTN BUS 10	-	-	-	165	165	165	nS	12
2	tpd	(F02) to DESTN BUS 9	-	-	-	165	165	165	nS	12
2	tpd	(F02) to DESTN BUS 8	-	-	-	165	165	165	nS	12
2	tpd	(F02) to DESTN BUS 7	-	-	-	165	165	165	nS	12
2	tpd	(F02) to DESTN BUS 6	-	-	-	165	165	165	nS	12
2	tpd	(F02) to DESTN BUS 5	-	-	-	165	165	165	nS	12
2	tpd	(F02) to DESTN BUS 4	-	-	-	165	165	165	nS	12
2	tpd	(F02) to DESTN BUS 3	-	-	-	165	165	165	nS	12
2	tpd	(F02) to DESTN BUS 2	-	-	-	165	165	165	nS	12
2	tpd	(F02) to DESTN BUS 1	-	-	-	165	165	165	nS	12
2	tpd	(F02) to DESTN BUS 0	-	-	-	165	165	165	nS	12
2	tpd	(F01) to DESTN BUS 15	-	-	-	165	165	165	nS	12
2	tpd	(F01) to DESTN BUS 14	-	-	-	165	165	165	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tpd	(F01) to DESTN BUS 13	-	-	-	165	165	165	nS	12
2	tpd	(F01) to DESTN BUS 12	-	-	-	165	165	165	nS	12
2	tpd	(F01) to DESTN BUS 11	-	-	-	165	165	165	nS	12
2	tpd	(F01) to DESTN BUS 10	-	-	-	165	165	165	nS	12
2	tpd	(F01) to DESTN BUS 9	-	-	-	165	165	165	nS	12
2	tpd	(F01) to DESTN BUS 8	-	-	-	165	165	165	nS	12
2	tpd	(F01) to DESTN BUS 7	-	-	-	165	165	165	nS	12
2	tpd	(F01) to DESTN BUS 6	-	-	-	165	165	165	nS	12
2	tpd	(F01) to DESTN BUS 5	-	-	-	165	165	165	nS	12
2	tpd	(F01) to DESTN BUS 4	-	-	-	165	165	165	nS	12
2	tpd	(F01) to DESTN BUS 3	-	-	-	165	165	165	nS	12
2	tpd	(F01) to DESTN BUS 2	-	-	-	165	165	165	nS	12
2	tpd	(F01) to DESTN BUS 1	-	-	-	165	165	165	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tpd	(F01) to DESTN BUS 0	-	-	-	165	165	165	nS	12
2	tpd	(SA03) to DESTN BUS 15	-	-	-	165	165	165	nS	12
2	tpd	(SA03) to DESTN BUS 14	-	-	-	165	165	165	nS	12
2	tpd	(SA03) to DESTN BUS 13	-	-	-	165	165	165	nS	12
2	tpd	(SA03) to DESTN BUS 12	-	-	-	165	165	165	nS	12
2	tpd	(SA03) to DESTN BUS 11	-	-	-	165	165	165	nS	12
2	tpd	(SA03) to DESTN BUS 10	-	-	-	165	165	165	nS	12
2	tpd	(SA03) to DESTN BUS 9	-	-	-	165	165	165	nS	12
2	tpd	(SA03) to DESTN BUS 8	-	-	-	165	165	165	nS	12
2	tpd	(SA03) to DESTN BUS 7	-	-	-	165	165	165	nS	12
2	tpd	(SA03) to DESTN BUS 6	-	-	-	165	165	165	nS	12
2	tpd	(SA03) to DESTN BUS 5	-	-	-	165	165	165	nS	12
2	tpd	(SA03) to DESTN BUS 4	-	-	-	165	165	165	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tpd	(SA03) to DESTN BUS 3	-	-	-	165	165	165	nS	12
2	tpd	(SA03) to DESTN BUS 2	-	-	-	165	165	165	nS	12
2	tpd	(SA03) to DESTN BUS 1	-	-	-	165	165	165	nS	12
2	tpd	(SA03) to DESTN BUS 0	-	-	-	165	165	165	nS	12
3	tpd	SOURCE A 15 to DESTN BUS 15	-	-	-	95	95	95	nS	12
3	tpd	SOURCE A 14 to DESTN BUS 14	-	-	-	95	95	95	nS	12
3	tpd	SOURCE A 13 to DESTN BUS 13	-	-	-	95	95	95	nS	12
3	tpd	SOURCE A 12 to DESTN BUS 12	-	-	-	95	95	95	nS	12
3	tpd	SOURCE A 11 to DESTN BUS 11	-	-	-	95	95	95	nS	12
3	tpd	SOURCE A 10 to DESTN BUS 10	-	-	-	95	95	95	nS	12
3	tpd	SOURCE A 9 to DESTN BUS 9	-	-	-	95	95	95	nS	12
3	tpd	SOURCE A 8 to DESTN BUS 8	-	-	-	95	95	95	nS	12
3	tpd	SOURCE A 7 to DESTN BUS 7	-	-	-	95	95	95	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
3	tpd	SOURCE A 6 to DESTN BUS 6	-	-	-	95	95	95	nS	12
3	tpd	SOURCE A 5 to DESTN BUS 5	-	-	-	95	95	95	nS	12
3	tpd	SOURCE A 4 to DESTN BUS 4	-	-	-	95	95	95	nS	12
3	tpd	SOURCE A 3 to DESTN BUS 3	-	-	-	95	95	95	nS	12
3	tpd	SOURCE A 2 to DESTN BUS 2	-	-	-	95	95	95	nS	12
3	tpd	SOURCE A 1 to DESTN BUS 1	-	-	-	95	95	95	nS	12
3	tpd	SOURCE A 0 to DESTN BUS 0	-	-	-	95	95	95	nS	12
4	tpd	TOGGLE ALU to DESTN BUS 15	-	-	-	105	105	105	nS	12
4	tpd	TOGGLE ALU to DESTN BUS 14	-	-	-	105	105	105	nS	12
4	tpd	TOGGLE ALU to DESTN BUS 13	-	-	-	105	105	105	nS	12
4	tpd	TOGGLE ALU to DESTN BUS 12	-	-	-	105	105	105	nS	12
4	tpd	TOGGLE ALU to DESTN BUS 11	-	-	-	105	105	105	nS	12
4	tpd	TOGGLE ALU to DESTN BUS 10	-	-	-	105	105	105	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tpd	TOGGLE ALU to DESTN BUS 9	-	-	-	105	105	105	nS	12
4	tpd	TOGGLE ALU to DESTN BUS 8	-	-	-	105	105	105	nS	12
4	tpd	TOGGLE ALU to DESTN BUS 7	-	-	-	105	105	105	nS	12
4	tpd	TOGGLE ALU to DESTN BUS 6	-	-	-	105	105	105	nS	12
4	tpd	TOGGLE ALU to DESTN BUS 5	-	-	-	105	105	105	nS	12
4	tpd	TOGGLE ALU to DESTN BUS 4	-	-	-	105	105	105	nS	12
4	tpd	TOGGLE ALU to DESTN BUS 3	-	-	-	105	105	105	nS	12
4	tpd	TOGGLE ALU to DESTN BUS 2	-	-	-	105	105	105	nS	12
4	tpd	TOGGLE ALU to DESTN BUS 1	-	-	-	105	105	105	nS	12
4	tpd	TOGGLE ALU to DESTN BUS 0	-	-	-	105	105	105	nS	12
5	tpd	DESTN BUS 15 to ALU ZERO	-	-	-	150	150	150	nS	12
5	tpd	DESTN BUS 14 to ALU ZERO	-	-	-	150	150	150	nS	12
5	tpd	DESTN BUS 13 to ALU ZERO	-	-	-	150	150	150	nS	12
5	tpd	DESTN BUS 12 to ALU ZERO	-	-	-	150	150	150	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
5	tpd	DESTN BUS 11 to ALU ZERO	-	-	-	150	150	150	nS	12
5	tpd	DESTN BUS 10 to ALU ZERO	-	-	-	150	150	150	nS	12
5	tpd	DESTN BUS 9 to ALU ZERO	-	-	-	150	150	150	nS	12
5	tpd	DESTN BUS 8 to ALU ZERO	-	-	-	150	150	150	nS	12
5	tpd	DESTN BUS 7 to ALU ZERO	-	-	-	150	150	150	nS	12
5	tpd	DESTN BUS 6 to ALU ZERO	-	-	-	150	150	150	nS	12
5	tpd	DESTN BUS 5 to ALU ZERO	-	-	-	150	150	150	nS	12
5	tpd	DESTN BUS 4 to ALU ZERO	-	-	-	150	150	150	nS	12
5	tpd	DESTN BUS 3 to ALU ZERO	-	-	-	150	150	150	nS	12
5	tpd	DESTN BUS 2 to ALU ZERO	-	-	-	150	150	150	nS	12
5	tpd	DESTN BUS 1 to ALU ZERO	-	-	-	150	150	150	nS	12
5	tpd	DESTN BUS 0 to ALU ZERO	-	-	-	150	150	150	nS	12
6	tpd	MICRO IR (S) to JMP TRUE	-	-	-	65	65	65	nS	12
7	tpd	BC/MC 03 to JMP TRUE	-	-	-	40	40	40	nS	12
7	tpd	BC/MC 00 to JMP TRUE	-	-	-	40	40	40	nS	12
8	tpd	SYS CLOCK to SERIAL OUT	-	-	-	60	60	60	nS	12
9	tsu	DESTN BUS 15 to DESTN CK	165	165	165	-	-	-	nS	12
9	tsu	DESTN BUS 14 to DESTN CK	165	165	165	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
9	tsu	DESTN BUS 13 to DESTN CK	165	165	165	-	-	-	nS	12
9	tsu	DESTN BUS 12 to DESTN CK	165	165	165	-	-	-	nS	12
9	tsu	DESTN BUS 11 to DESTN CK	165	165	165	-	-	-	nS	12
9	tsu	DESTN BUS 10 to DESTN CK	165	165	165	-	-	-	nS	12
9	tsu	DESTN BUS 9 to DESTN CK	165	165	165	-	-	-	nS	12
9	tsu	DESTN BUS 8 to DESTN CK	165	165	165	-	-	-	nS	12
9	tsu	DESTN BUS 7 to DESTN CK	165	165	165	-	-	-	nS	12
9	tsu	DESTN BUS 6 to DESTN CK	165	165	165	-	-	-	nS	12
9	tsu	DESTN BUS 5 to DESTN CK	165	165	165	-	-	-	nS	12
9	tsu	DESTN BUS 4 to DESTN CK	165	165	165	-	-	-	nS	12
9	tsu	DESTN BUS 3 to DESTN CK	165	165	165	-	-	-	nS	12
9	tsu	DESTN BUS 2 to DESTN CK	165	165	165	-	-	-	nS	12
9	tsu	DESTN BUS 1 to DESTN CK	165	165	165	-	-	-	nS	12
9	tsu	DESTN BUS 0 to DESTN CK	165	165	165	-	-	-	nS	12
10	tsu	DESTN BUS 15 to MICRO IR (S)	165	165	165	-	-	-	nS	12
10	tsu	DESTN BUS 14 to MICRO IR (S)	165	165	165	-	-	-	nS	12
10	tsu	DESTN BUS 13 to MICRO IR (S)	165	165	165	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
10	tsu	DESTN BUS 12 to MICRO IR (S)	165	165	165	-	-	-	nS	12
10	tsu	DESTN BUS 11 to MICRO IR (S)	165	165	165	-	-	-	nS	12
10	tsu	DESTN BUS 10 to MICRO IR (S)	165	165	165	-	-	-	nS	12
10	tsu	DESTN BUS 9 to MICRO IR (S)	165	165	165	-	-	-	nS	12
10	tsu	DESTN BUS 8 to MICRO IR (S)	165	165	165	-	-	-	nS	12
10	tsu	DESTN BUS 7 to MICRO IR (S)	165	165	165	-	-	-	nS	12
10	tsu	DESTN BUS 6 to MICRO IR (S)	165	165	165	-	-	-	nS	12
10	tsu	DESTN BUS 5 to MICRO IR (S)	165	165	165	-	-	-	nS	12
10	tsu	DESTN BUS 4 to MICRO IR (S)	165	165	165	-	-	-	nS	12
10	tsu	DESTN BUS 3 to MICRO IR (S)	165	165	165	-	-	-	nS	12
10	tsu	DESTN BUS 2 to MICRO IR (S)	165	165	165	-	-	-	nS	12
10	tsu	DESTN BUS 1 to MICRO IR (S)	165	165	165	-	-	-	nS	12
10	tsu	DESTN BUS 0 to MICRO IR (S)	165	165	165	-	-	-	nS	12

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6.4 Module function. The Arithmetic Logic Unit (ALU) module contains a 16-bit microprocessor that performs the Central Processor (CP) arithmetic and logic functions, a clock generator that provides CP timing, and a serial maintenance interface for Built-In-Test (BIT), control, and display functions. The clock generator is a multiple rate clock generator that provides four timing pulses each microcycle (one microinstruction is executed each microcycle). The timing pulses are generated from a 20 MHz oscillator input, and the length of the microcycle is determined by each microinstruction. This permits varying the length of the microcycle to fit the operation. Microcycle lengths are 200, 250, 300, and 350 nanoseconds. The clock generator may be stopped between microcycles for some operations (for example: to permit completion of a memory reference). The clock generator outputs are sent to the system clock for distribution to the CP modules. The microinstruction determines the function to perform, the source of the operands, and where to place the result. The principle CP data transfer path is the source B/destination bus. The source B/destination bus is used to transfer both the source data and the destination data during a microcycle. The source A bus provides another source data path for operands from a general register, multiplier/accumulator, or the shift matrix. The 24-bit Serial Maintenance Interface (SMI) register is used by the maintenance processor for BIT fault isolation and for control and display functions. The register holds display data from the source B/destination bus and miscellaneous control bits. The register is loaded or read serially by the maintenance processor.

6.4.1 Clock generator. The clock generator is a programmable multiple rate clock generator that generates the CP timing pulses. Timing pulses are generated from the 20 MHz oscillator input. The clock rate is determined by the microinstruction CLK field (bits 47 and 46) as follows:

(a) CLK = 0, rate = 200 nanoseconds.

(b) CLK = 1, rate = 350 nanoseconds.

(c) CLK = 2, rate = 250 nanoseconds.

(d) CLK = 3, rate = 300 nanoseconds.

The bus clock output tracks the 20 MHz oscillator input.

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6.4.2 Microprocessor and carry look-ahead. The microprocessor contains 16 16-bit internal working registers (located in a Random Access Memory (RAM)), an arithmetic logic unit and the associated shifting, decoding, and multiplexing circuits. The RAM has two address ports and two output ports. The ALU performs seven arithmetic and nine logic operations on two 16-bit operands. The microprocessor consists of four 4-bit integrated circuits. The carry look-ahead generator provides high speed carries across the four microprocessor devices. The microinstruction F field controls ALU operations. The microinstruction X, Source A (SA), and Source B (SB) fields determine operand sources and the D field determines where the result is stored. The X and SA fields select one operand from either the internal RAM or an external register via the source A bus (X = 0, selects internal RAM). The other operand is selected from the internal Q register (when F = 1-5, or 9-F), or by the SB field (when F = 0, or 6-8) from the internal RAM or an external register via the source B latch. Address D contains the macro-program counter and address F the macrostatus register 2. The remaining locations are used as micro-scratchpad registers. The SA field only addresses locations 0 through 7. When D field bit 4 equals one, the internal RAM is loaded from the ALU output at the end of the timing cycle. D field bits 3-0 address the register.

6.4.3 B address selector. The 4-bit selector determines the source for the microprocessor B address port. The Source/Destination (S/D) clock selects the source. When the clock is high, microinstruction bits 19-16 (SB3-SB0) are selected. When the clock is low, microinstruction bits 27-24 (D3-D0) are selected.

6.4.4 Source B latch. The 16-bit latch holds the operand used by the microprocessor from an external register. The operand is latched from the source B/destination bus. The output is enabled and placed on the microprocessor B data port bus by a microinstruction with SB4 = 1.

6.4.5 Microaccumulator source buffer. The 16-bit buffer holds microaccumulator data (from microprocessor RAM) which allows placing the data on the source B/destination bus during source time. The buffer output is enabled by a microinstruction with SB4 = 0.

6.4.6 ALU status register. The 4-bit register holds the following ALU status bits: zero, negative, carry, and overflow. The register is loaded by a microinstruction with S = 1.

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6.4.7 Microbranch control. Microbranch control determines if the branch condition specified by the microinstruction BC field is true by interrogating the ALU status register. The status register is interrogated for BC = 0 through 7.

6.4.8 Control circuits. The control circuits consist of a control Programmable Array Logic (PAL) and miscellaneous circuits that control ALU and Serial Maintenance Interface (SMI) operations. The control PAL decodes the microinstruction D, SB, and microcontrol instruction type fields.

6.4.9 Serial Maintenance Interface (SMI) circuits. The SMI circuits consist of the SMI display register, serial control register, parallel control register, and SMI sync register. The 16-bit display register and 8-bit serial control register hold the ALU SMI word. The word consists of display data and miscellaneous controls. The display register holds the display data and the serial control register holds the miscellaneous controls. The display register is parallel read by a microinstruction with SB = 1F and parallel loaded by a microinstruction with D = 0F via the source B/destination bus. The serial control register is parallel loaded from the parallel control register by the Load signal and maintenance clock from the control and maintenance interface. When the maintenance processor enables ALU SMI, the display and serial control registers are right shifted one bit position on each maintenance processor clock pulse. The data is transferred to/from the maintenance processor via the serial maintenance interface on the Bit Electronics module. The 8-bit parallel control register holds the SMI controls to provide isolation from the serial data path. The register is loaded from the serial control register when the SMI word bit 7 equals one and the Load Enable signal and maintenance clock are present. Bits 0 and 2 can also be set by the CP. All control bits are cleared by a Master Clear. The SMI sync register provides synchronization between the maintenance processor clock and CP clocks to stabilize the MP to MPC Interrupt and MPC Hold signals.

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6.4.10 Description of I/O pins.

Signal name	Pin number	I/O	Description
DESTN BUS 15	9	I/O	SCE B/DESTN BUS 15
DESTN BUS 14	11	I/O	SCE B/DESTN BUS 14
DESTN BUS 13	12	I/O	SCE B/DESTN BUS 13
DESTN BUS 12	13	I/O	SCE B/DESTN BUS 12
DESTN BUS 11	15	I/O	SCE B/DESTN BUS 11
DESTN BUS 10	16	I/O	SCE B/DESTN BUS 10
DESTN BUS 9	17	I/O	SCE B/DESTN BUS 9
DESTN BUS 8	18	I/O	SCE B/DESTN BUS 8
DESTN BUS 7	87	I/O	SCE B/DESTN BUS 7
DESTN BUS 6	88	I/O	SCE B/DESTN BUS 6
DESTN BUS 5	89	I/O	SCE B/DESTN BUS 5
DESTN BUS 4	91	I/O	SCE B/DESTN BUS 4
DESTN BUS 3	93	I/O	SCE B/DESTN BUS 3
DESTN BUS 2	94	I/O	SCE B/DESTN BUS 2
DESTN BUS 1	95	I/O	SCE B/DESTN BUS 1
DESTN BUS 0	96	I/O	SCE B/DESTN BUS 0
SOURCE A 15	2	I	SOURCE A 15
SOURCE A 14	3	I	SOURCE A 14
SOURCE A 13	4	I	SOURCE A 13
SOURCE A 12	5	I	SOURCE A 12

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6.4.10 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
BC/MC 03	19	I	MICRO IR (BC/MC 03)
BC/MC 02	20	I	MICRO IR (BC/MC 02)
BC/MC 01	21	I	MICRO IR (BC/MC 01)
BC/MC 00	22	I	MICRO IR (BC/MC 00)
EX BRAN COND	23	I	EXPANDED BRANCH CONDITION
MASTER CLEAR	24	I	MASTER CLEAR
(SB04)	26	I	MICRO IR (SB04)
MC CL FORMAT	27	I	MICRO CONTROL FORMAT
MC IR (F00)	28	I	MICRO IR (F00)
TOGGLE ALU	29	I	TOGGLE ALU FUNCTION
MCU (00)	30	I	MICRO IR (MCU 00)
MCU (01)	32	I	MICRO IR (MCU 01)
(D00)	33	I	MICRO IR (D00)
(SB03)	35	I	MICRO IR (SB 03)
(SB02)	36	I	MICRO IR (SB 02)
(SB01)	37	I	MICRO IR (SB 01)
(SB00)	38	I	MICRO IR (SB 00)
(D01)	39	I	MICRO IR (D01)
(F03)	41	I	MICRO IR (F03)
(F02)	42	I	MICRO IR (F02)

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6.4.10 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
(F01)	43	I	MICRO IR (F01)
(SA02)	44	I	MICRO IR (SA 02)
(SA01)	45	I	MICRO IR (SA 01)
(SA00)	46	I	MICRO IR (SA 00)
SOURCE A 11	47	I	SOURCE A 11
SOURCE A 10	48	I	SOURCE A 10
SOURCE A 9	49	I	SOURCE A 9
SOURCE A 8	50	I	SOURCE A 8
SOURCE A 3	52	I	SOURCE A 3
SOURCE A 2	53	I	SOURCE A 2
SOURCE A 1	54	I	SOURCE A 1
SOURCE A 0	55	I	SOURCE A 0
CLK RATE 2	56	I	CLK RATE CONT 2
CLK RATE 1	57	I	CLK RATE CONT 1
CLK RATE 0	58	I	CLK RATE CONT 0
20 MHZ OSC	63	I	20 MHz OSCILLATOR (IN)
DESTN CK	64	I	SCE B/DESTN CLK
(SA03)	66	I	MICRO IR (SA 03)
B BUS ENABLE	67	I	SOURCE B BUS ENABLE
WAIT CLOCK	69	I	WAIT CONT CLK (CK IN)

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6.4.10 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
WAIT REQUEST	71	I	WAIT REQUEST
READY	72	I	READY
SMI DATA OUT	76	I	SMI DATA OUT (ALU IN)
MICRO IR (S)	77	I	MICRO IR (S)
SMI LD/SF CT	78	I	SMI LOAD/SHIFT CONT
DISABLE ALU	79	I	DISABLE ALU 3-STATE
(D02)	80	I	MICRO IR (D02)
(D03)	82	I	MICRO IR (D03)
(D04)	83	I	MICRO IR (D04)
ENABLE CHAIN	84	I	SMI ENABLE CHAIN
(C0)	85	I	MICRO IR (C0)
(C1)	86	I	MICRO IR (C1)
SYS CLOCK	92	I	MP SYSTEM CLOCK
SOURCE A 7	97	I	SOURCE A 7
SOURCE A 6	98	I	SOURCE A 6
SOURCE A 5	99	I	SOURCE A 5
SOURCE A 4	100	I	SOURCE A 4
ALU OVL	6	O	ALU OVERFLOW
ALU CARRY	7	O	ALU CARRY
ALU NEG	8	O	ALU NEGATIVE

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6.4.10 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
JMP TRUE	14	O	JUMP CONDITION TRUE
ALU ZERO	34	O	ALU ZERO
20 MHZ CLOCK	59	O	20 MHz CLOCK (F0 CLOCK)
ALU C1 CLOCK	61	O	ALU C1 CLOCK
MP TO MPC	62	O	MP TO MPC INTERRUPT
ALU C3 CLOCK	65	O	ALU C3 CLOCK (PROM ENBL CLK)
ALU C2 CLOCK	68	O	ALU C2 CLOCK
ALU C4 CLOCK	70	O	ALU C4 CLOCK
WAIT ACK	73	O	WAIT ACK
SERIAL OUT	74	O	SERIAL OUT

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6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Preparing activity:
Navy - SH

(Project 5963-N716)

Agent:
NWSC Crane

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TABLE II. Isolation test.

Test cond	Test description	Connect		25 ± 5°C Initial limits		Units	Notes
		A to pin	B to pin	Min	Max		
A	Isolation, frame	C	Frame	10	INF	Mohm	1-8

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A
Isolation measurement	NA	± 500 KOhm	A

2. The ohmmeter used shall be item R06.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, and 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

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NOTES: - Continued.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

7. The test circuit shall be as indicated on figure 6.

8. Abbreviations used are defined in 6.1.

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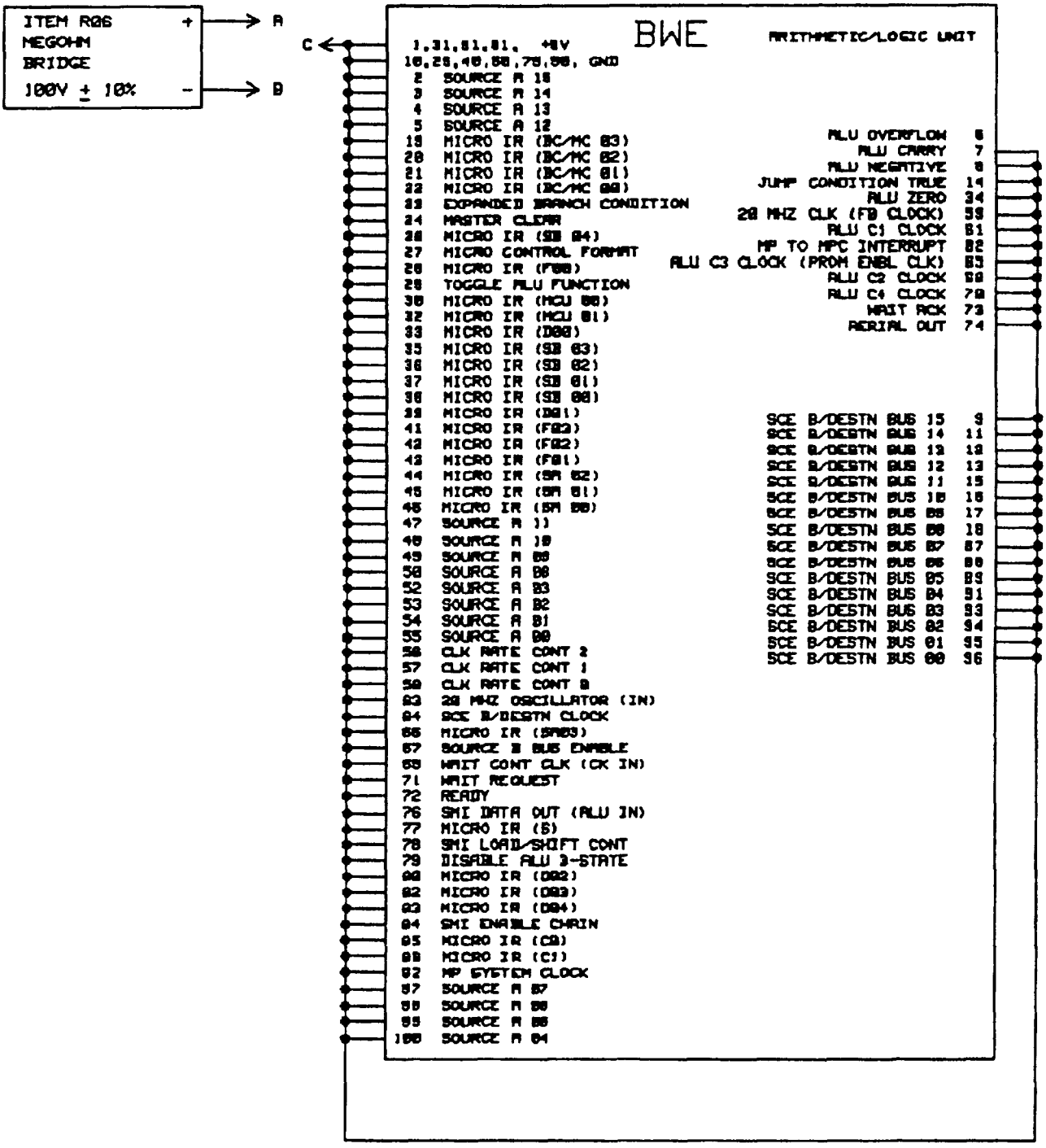


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect		25 ± 5°C Initial limits		Units	Notes
		A to pin	B to pin	Min	Max		
B1	Continuity, GND	10	25	0	1	Ohm	1-5
B2	Continuity, GND	10	40				
B3	Continuity, GND	60	75				
B4	Continuity, GND	60	90				
B5	Continuity, Vcc	1	31				
B6	Continuity, Vcc	51	81	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B6

2. The ohmmeter used shall be item R07.

3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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BWE — ARITHMETIC/LOGIC UNIT

1,31,51,81,	+5V		
10,25,40,50,75,90,	GND		
2	SOURCE A I5		
3	SOURCE A I4		
4	SOURCE A I3		
5	SOURCE A I2		
19	MICRO IR (BC/MC 02)	ALU OVERFLOW	6
20	MICRO IR (BC/MC 02)	ALU CARRY	7
21	MICRO IR (BC/MC 01)	ALU NEGATIVE	8
22	MICRO IR (BC/MC 00)	JUMP CONDITION TRUE	14
23	EXPANDED BRANCH CONDITION	ALU ZERO	34
24	MASTER CLEAR	20 MHZ CLK (F8 CLOCK)	58
26	MICRO IR (SB 04)	ALU C1 CLOCK	61
27	MICRO CONTROL FORMAT	MP TO MPC INTERRUPT	62
28	MICRO IR (F00)	ALU C3 CLOCK (FROM ENBL CLK)	65
29	TOGGLE ALU FUNCTION	ALU C2 CLOCK	69
30	MICRO IR (MCU 03)	ALU C4 CLOCK	70
32	MICRO IR (MCU 01)	WAIT ACK	73
33	MICRO IR (D00)	AERIAL OUT	74
35	MICRO IR (SB 02)		
36	MICRO IR (SB 02)		
37	MICRO IR (SB 01)		
38	MICRO IR (SB 00)		
39	MICRO IR (D01)	SCE B/DESTN BUS I5	9
41	MICRO IR (F03)	SCE B/DESTN BUS I4	11
42	MICRO IR (F02)	SCE B/DESTN BUS I3	12
43	MICRO IR (F01)	SCE B/DESTN BUS I2	13
44	MICRO IR (SA 02)	SCE B/DESTN BUS I1	15
45	MICRO IR (SA 01)	SCE B/DESTN BUS I0	16
46	MICRO IR (SA 00)	SCE B/DESTN BUS 09	17
47	SOURCE A I1	SCE B/DESTN BUS 08	18
48	SOURCE A I0	SCE B/DESTN BUS 07	87
49	SOURCE A 09	SCE B/DESTN BUS 06	88
50	SOURCE A 08	SCE B/DESTN BUS 05	89
52	SOURCE A 03	SCE B/DESTN BUS 04	91
53	SOURCE A 02	SCE B/DESTN BUS 03	93
54	SOURCE A 01	SCE B/DESTN BUS 02	94
55	SOURCE A 00	SCE B/DESTN BUS 01	95
56	CLK RATE CONT 2	SCE B/DESTN BUS 00	96
57	CLK RATE CONT 1		
58	CLK RATE CONT 0		
63	20 MHZ OSCILLATOR (IN)		
64	SCE B/DESTN CLOCK		
66	MICRO IR (SA03)		
67	SOURCE B BUS ENABLE		
68	WAIT CONT CLK (CK IN)		
71	WAIT REQUEST		
72	READY		
76	SMI DATA OUT (ALU IN)		
77	MICRO IR (S)		
78	SMI LOAD/SHIFT CONT		
79	DISABLE ALU 3-STATE		
80	MICRO IR (D02)		
82	MICRO IR (D03)		
83	MICRO IR (D04)		
84	SMI ENABLE CHAIN		
85	MICRO IR (C0)		
86	MICRO IR (C1)		
92	MP SYSTEM CLOCK		
97	SOURCE A 07		
98	SOURCE A 06		
99	SOURCE A 05		
100	SOURCE A 04		

FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	3.67 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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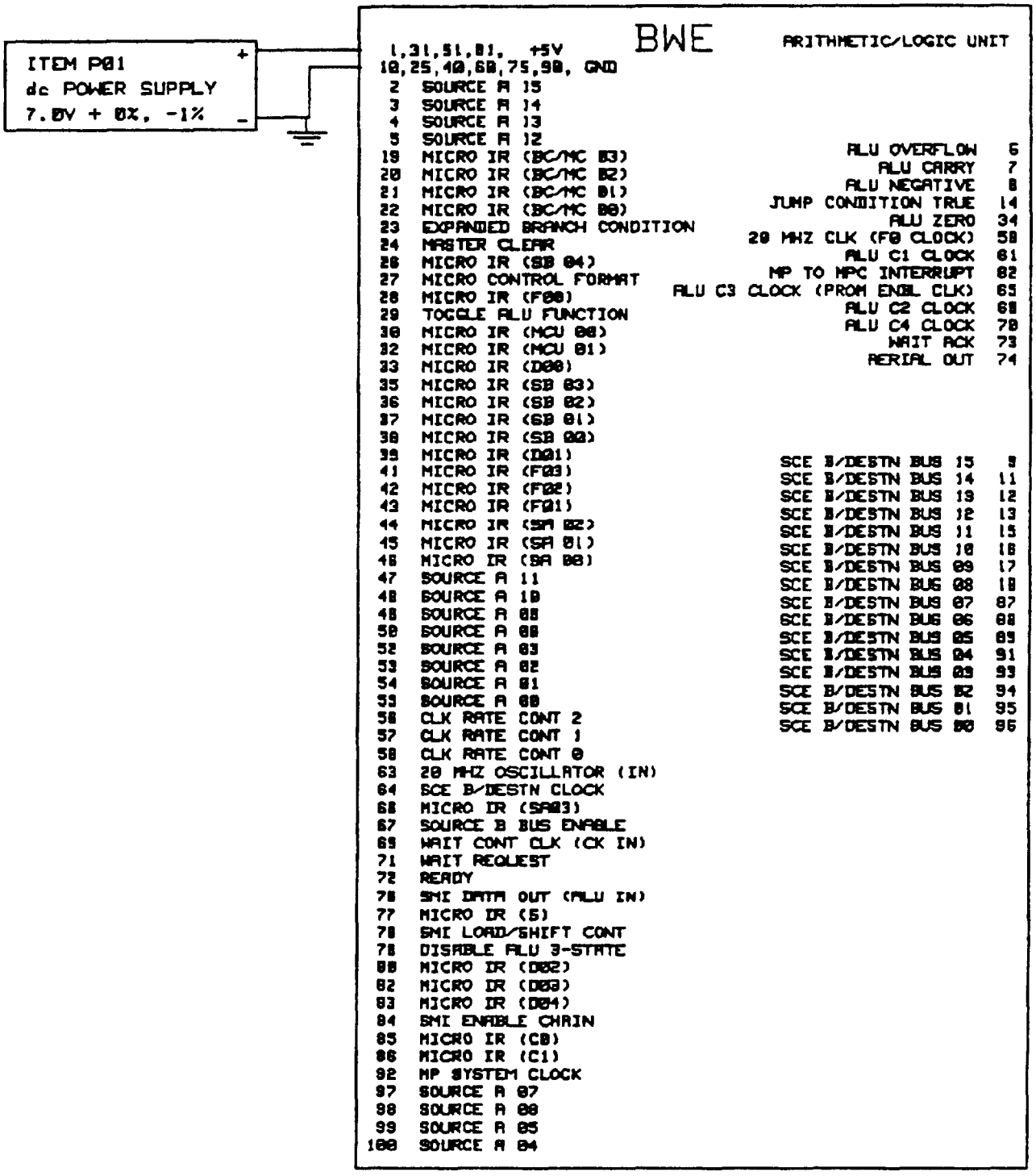


FIGURE 8. Vcc stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	SOURCE A 15	2	5.5	0.0	1.0	mA	1-3
D2	SOURCE A 14	3					
D3	SOURCE A 13	4					
D4	SOURCE A 12	5	5.5	0.0	1.0	mA	1-3
D5	BC/MC 03	19	5.5	0.0	2.0	mA	1-3
D6	BC/MC 02	20	5.5	0.0	1.1	mA	1-3
D7	BC/MC 01	21					
D8	BC/MC 00	22	5.5	0.0	1.1	mA	1-3
D9	EX BRAN COND	23	5.5	0.0	1.0	mA	1-3
D10	MASTER CLEAR	24	5.5	0.0	0.1	mA	1-3
D11	(SB04)	26	5.5	0.0	0.2	mA	1-3
D12	MC CL FORMAT	27	5.5	0.0	1.0	mA	1-3
D13	MC IR (F00)	28	5.5	0.0	0.1	mA	1-3
D14	TOGGLE ALU	29	5.5	0.0	0.2	mA	1-3
D15	MCU (00)	30	5.5	0.0	1.0	mA	1-3
D16	MCU (01)	32	5.5	0.0	1.0	mA	1-3
D17	(D00)	33	5.5	0.0	1.1	mA	1-3
D18	(SB03)	35	5.5	0.0	0.2	mA	1-3
D19	(SB02)	36					
D20	(SB01)	37	5.5	0.0	0.2	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
D21	(SB00)	38	5.5	0.0	0.2	mA	1-3
D22	(D01)	39	5.5	0.0	1.1	mA	1-3
D23	(F03)	41	5.5	0.0	4.1	mA	1-3
D24	(F02)	42					
D25	(F01)	43	5.5	0.0	4.1	mA	1-3
D26	(SA02)	44	5.5	0.0	4.0	mA	1-3
D27	(SA01)	45					
D28	(SA00)	46	5.5	0.0	4.0	mA	1-3
D29	SOURCE A 11	47	5.5	0.0	1.0	mA	1-3
D30	SOURCE A 10	48					
D31	SOURCE A 9	49					
D32	SOURCE A 8	50					
D33	SOURCE A 3	52					
D34	SOURCE A 2	53					
D35	SOURCE A 1	54					
D36	SOURCE A 0	55	5.5	0.0	1.0	mA	1-3
D37	CLK RATE 2	56	5.5	0.0	0.1	mA	1-3
D38	CLK RATE 1	57					
D39	CLK RATE 0	58	5.5	0.0	0.1	mA	1-3
D40	20 MHZ OSC	63	5.5	0.0	1.0	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D41	DESTN CK	64	5.5	0.0	2.0	mA	1-3
D42	(SA03)	66	5.5	0.0	4.0	mA	1-3
D43	B BUS ENABLE	67	5.5	0.0	2.0	mA	1-3
D44	WAIT CLOCK	69	5.5	0.0	1.0	mA	1-3
D45	WAIT REQUEST	71	5.5	0.0	1.0	mA	1-3
D46	READY	72	5.5	0.0	0.1	mA	1-3
D47	SMI DATA OUT	76					
D48	MICRO IR (S)	77	5.5	0.0	0.1	mA	1-3
D49	SMI LD/SF CT	78	5.5	0.0	2.0	mA	1-3
D50	DISABLE ALU	79	5.5	0.0	1.0	mA	1-3
D51	(D02)	80	5.5	0.0	1.1	mA	1-3
D52	(D03)	82					
D53	(D04)	83	5.5	0.0	1.1	mA	1-3
D54	ENABLE CHAIN	84	5.5	-0.02	3.02	mA	1-3
D55	(C0)	85	5.5	0.0	1.1	mA	1-3
D56	(C1)	86	5.5	0.0	1.2	mA	1-3
D57	SYS CLOCK	92	5.5	0.0	1.0	mA	1-3
D58	SOURCE A 7	97					
D59	SOURCE A 6	98					
D60	SOURCE A 5	99					
D61	SOURCE A 4	100	5.5	0.0	1.0	mA	1-3

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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D61
Power supply current limit	3.06 A	NA	D1-D61
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D61
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 D1-D61 D1-D61
Input current measurement	NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA	± 50 uA ± 100 uA ± 55 uA ± 50 uA ± 5 uA ± 10 uA ± 50 uA ± 5 uA ± 10 uA ± 50 uA ± 55 uA ± 10 uA ± 55 uA ± 205 uA ± 200 uA ± 50 uA ± 5 uA ± 50 uA ± 100 uA ± 200 uA ± 100 uA ± 50 uA	D1-D4 D5 D6-D8 D9 D10 D11 D12 D13 D14 D15,D16 D17 D18-D21 D22 D23-D25 D26-D28 D29-D36 D37-D39 D40 D41 D42 D43 D44,D45

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 5 uA	D46-D48
	NA	± 100 uA	D49
	NA	± 50 uA	D50
	NA	± 55 uA	D51-D53
	NA	± 152 uA	D54
	NA	± 55 uA	D55
	NA	± 60 uA	D56
	NA	± 50 uA	D57-D61

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	ALU OVL	6	0.0	-150.0	-60.0	mA	1-4
E2	ALU CARRY	7					
E3	ALU NEG	8	0.0	-150.0	-60.0	mA	1-4
E4	DESTN BUS 15	9	0.0	-85.0	-30.0	mA	1-4
E5	DESTN BUS 15	9	0.0	-130.0	-15.0	mA	1-4
E6	DESTN BUS 15	9	0.0	-220.0	-30.0	mA	1-4
E7	DESTN BUS 14	11	0.0	-85.0	-30.0	mA	1-4
E8	DESTN BUS 14	11	0.0	-130.0	-15.0	mA	1-4
E9	DESTN BUS 14	11	0.0	-220.0	-30.0	mA	1-4
E10	DESTN BUS 13	12	0.0	-85.0	-30.0	mA	1-4
E11	DESTN BUS 13	12	0.0	-130.0	-15.0	mA	1-4
E12	DESTN BUS 13	12	0.0	-220.0	-30.0	mA	1-4
E13	DESTN BUS 12	13	0.0	-130.0	-15.0	mA	1-4
E14	DESTN BUS 12	13	0.0	-220.0	-30.0	mA	1-4
E15	DESTN BUS 12	13	0.0	-85.0	-30.0	mA	1-4
E16	JMP TRUE	14	0.0	-150.0	-60.0	mA	1-4
E17	DESTN BUS 11	15	0.0	-85.0	-30.0	mA	1-4
E18	DESTN BUS 11	15	0.0	-130.0	-15.0	mA	1-4
E19	DESTN BUS 11	15	0.0	-220.0	-30.0	mA	1-4
E20	DESTN BUS 10	16	0.0	-85.0	-30.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E21	DESTN BUS 10	16	0.0	-130.0	-15.0	mA	1-4
E22	DESTN BUS 10	16	0.0	-220.0	-30.0	mA	1-4
E23	DESTN BUS 9	17	0.0	-85.0	-30.0	mA	1-4
E24	DESTN BUS 9	17	0.0	-130.0	-15.0	mA	1-4
E25	DESTN BUS 9	17	0.0	-220.0	-30.0	mA	1-4
E26	DESTN BUS 8	18	0.0	-85.0	-30.0	mA	1-4
E27	DESTN BUS 8	18	0.0	-130.0	-15.0	mA	1-4
E28	DESTN BUS 8	18	0.0	-220.0	-30.0	mA	1-4
E29	ALU ZERO	34	0.0	-100.0	-40.0	mA	1-4
E30	20 MHZ CLOCK	59	0.0	-85.0	-30.0	mA	1-4
E31	ALU C1 CLOCK	61	0.0	-85.0	-30.0	mA	1-4
E32	MP TO MPC	62	0.0	-100.0	-20.0	mA	1-4
E33	ALU C3 CLOCK	65	0.0	-85.0	-30.0	mA	1-4
E34	ALU C2 CLOCK	68					
E35	ALU C4 CLOCK	70					
E36	WAIT ACK	73	0.0	-85.0	-30.0	mA	1-4
E37	SERIAL OUT	74	0.0	-120.0	-30.0	mA	1-4
E38	DESTN BUS 7	87	0.0	-85.0	-30.0	mA	1-4
E39	DESTN BUS 7	87	0.0	-130.0	-15.0	mA	1-4
E40	DESTN BUS 7	87	0.0	-220.0	-30.0	mA	1-4
E41	DESTN BUS 6	88	0.0	-85.0	-30.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E42	DESTN BUS 6	88	0.0	-130.0	-15.0	mA	1-4
E43	DESTN BUS 6	88	0.0	-220.0	-30.0	mA	1-4
E44	DESTN BUS 5	89	0.0	-85.0	-30.0	mA	1-4
E45	DESTN BUS 5	89	0.0	-130.0	-15.0	mA	1-4
E46	DESTN BUS 5	89	0.0	-220.0	-30.0	mA	1-4
E47	DESTN BUS 4	91	0.0	-85.0	-30.0	mA	1-4
E48	DESTN BUS 4	91	0.0	-130.0	-15.0	mA	1-4
E49	DESTN BUS 4	91	0.0	-220.0	-30.0	mA	1-4
E50	DESTN BUS 3	93	0.0	-85.0	-30.0	mA	1-4
E51	DESTN BUS 3	93	0.0	-130.0	-15.0	mA	1-4
E52	DESTN BUS 3	93	0.0	-240.0	-30.0	mA	1-4
E53	DESTN BUS 2	94	0.0	-85.0	-30.0	mA	1-4
E54	DESTN BUS 2	94	0.0	-130.0	-15.0	mA	1-4
E55	DESTN BUS 2	94	0.0	-220.0	-30.0	mA	1-4
E56	DESTN BUS 1	95	0.0	-85.0	-30.0	mA	1-4
E57	DESTN BUS 1	95	0.0	-130.0	-15.0	mA	1-4
E58	DESTN BUS 1	95	0.0	-220.0	-30.0	mA	1-4
E59	DESTN BUS 0	96	0.0	-85.0	-30.0	mA	1-4
E60	DESTN BUS 0	96	0.0	-130.0	-15.0	mA	1-4
E61	DESTN BUS 0	96	0.0	-220.0	-30.0	mA	1-4

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E61
Power supply current limit	3.06 A	NA	E1-E61
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E61
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	E1-E61
low, (VL)	0.0 V	± 100 mV	E1-E61
Output current measurement	NA	± 4.5 mA	E1-E3
	NA	± 2.75 mA	E4
	NA	± 5.75 mA	E5
	NA	± 9.50 mA	E6
	NA	± 2.75 mA	E7
	NA	± 5.75 mA	E8
	NA	± 9.50 mA	E9
	NA	± 2.75 mA	E10
	NA	± 5.75 mA	E11
	NA	± 9.50 mA	E12
	NA	± 5.75 mA	E13
	NA	± 9.50 mA	E14
	NA	± 2.75 mA	E15
	NA	± 4.50 mA	E16
	NA	± 2.75 mA	E17
	NA	± 5.75 mA	E18
	NA	± 9.50 mA	E19
	NA	± 2.75 mA	E20
	NA	± 5.75 mA	E21
	NA	± 9.50 mA	E22
	NA	± 2.75 mA	E23

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Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output current measurement	NA	± 5.75 mA	E24
	NA	± 9.50 mA	E25
	NA	± 2.75 mA	E26
	NA	± 5.75 mA	E27
	NA	± 9.50 mA	E28
	NA	± 3.0 mA	E29
	NA	± 2.75 mA	E30,E31
	NA	± 4.0 mA	E32
	NA	± 2.75 mA	E33-E36
	NA	± 4.50 mA	E37
	NA	± 2.75 mA	E38
	NA	± 5.75 mA	E39
	NA	± 9.50 mA	E40
	NA	± 2.75 mA	E41
	NA	± 5.75 mA	E42
	NA	± 9.50 mA	E43
	NA	± 2.75 mA	E44
	NA	± 5.75 mA	E45
	NA	± 9.50 mA	E46
	NA	± 2.75 mA	E47
	NA	± 5.75 mA	E48
	NA	± 9.50 mA	E49
	NA	± 2.75 mA	E50
	NA	± 5.75 mA	E51
	NA	± 10.50 mA	E52
	NA	± 2.75 mA	E53
	NA	± 5.75 mA	E54
	NA	± 9.50 mA	E55
	NA	± 2.75 mA	E56
	NA	± 5.75 mA	E57
	NA	± 9.50 mA	E58
	NA	± 2.75 mA	E59
	NA	± 5.75 mA	E60
	NA	± 9.50 mA	E61

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	SOURCE A 15	2	2.4	-0.72	0.04	mA	1-3
F2	SOURCE A 14	3					
F3	SOURCE A 13	4					
F4	SOURCE A 12	5	2.4	-0.72	0.04	mA	1-3
F5	BC/MC 03	19	2.4	-2.25	0.075	mA	1-3
F6	BC/MC 02	20	2.4	-0.85	0.045	mA	1-3
F7	BC/MC 01	21					
F8	BC/MC 00	22	2.4	-0.85	0.045	mA	1-3
F9	EX BRAN COND	23	2.4	-2.0	0.05	mA	1-3
F10	MASTER CLEAR	24	2.4	-0.6	0.02	mA	1-3
F11	(SB04)	26	2.4	-1.0	0.04	mA	1-3
F12	MC CL FORMAT	27	2.4	-0.25	0.025	mA	1-3
F13	MC IR (F00)	28	2.4	-0.6	0.02	mA	1-3
F14	TOGGLE ALU	29	2.4	-1.2	0.04	mA	1-3
F15	MCU (00)	30	2.4	-0.25	0.025	mA	1-3
F16	MCU (01)	32	2.4	-0.25	0.025	mA	1-3
F17	(D00)	33	2.4	-0.85	0.045	mA	1-3
F18	(SB03)	35	2.4	-1.0	0.04	mA	1-3
F19	(SB02)	36					
F20	(SB01)	37	2.4	-1.0	0.04	mA	1-3

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Check the source to verify that this is the current version before use.

TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F21	(SB00)	38	2.4	-1.0	0.04	mA	1-3
F22	(D01)	39	2.4	-0.85	0.045	mA	1-3
F23	(F03)	41	2.4	-3.48	0.18	mA	1-3
F24	(F02)	42					
F25	(F01)	43	2.4	-3.48	0.18	mA	1-3
F26	(SA02)	44	2.4	-1.44	0.08	mA	1-3
F27	(SA01)	45					
F28	(SA00)	46	2.4	-1.44	0.08	mA	1-3
F29	SOURCE A 11	47	2.4	-0.72	0.04	mA	1-3
F30	SOURCE A 10	48					
F31	SOURCE A 9	49					
F32	SOURCE A 8	50					
F33	SOURCE A 3	52					
F34	SOURCE A 2	53					
F35	SOURCE A 1	54					
F36	SOURCE A 0	55	2.4	-0.72	0.04	mA	1-3
F37	CLK RATE 2	56	2.4	-0.4	0.02	mA	1-3
F38	CLK RATE 1	57					
F39	CLK RATE 0	58	2.4	-0.4	0.02	mA	1-3
F40	20 MHZ OSC	63	2.4	-2.0	0.05	mA	1-3

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Check the source to verify that this is the current version before use.

TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F41	DESTN CK	64	2.4	-4.0	0.1	mA	1-3
F42	(SA03)	66	2.4	-1.44	0.08	mA	1-3
F43	B BUS ENABLE	67	2.4	-4.0	0.1	mA	1-3
F44	WAIT CLOCK	69	2.4	-1.2	0.07	mA	1-3
F45	WAIT REQUEST	71	2.4	-0.8	0.05	mA	1-3
F46	READY	72	2.4	-0.4	0.02	mA	1-3
F47	SMI DATA OUT	76	2.4	-0.4	0.02	mA	1-3
F48	MICRO IR (S)	77	2.4	-0.6	0.02	mA	1-3
F49	SMI LD/SF CT	78	2.4	-4.0	0.1	mA	1-3
F50	DISABLE ALU	79	2.4	-2.0	0.05	mA	1-3
F51	(D02)	80	2.4	-0.85	0.045	mA	1-3
F52	(D03)	82	|	|	|	|	|
F53	(D04)	83	2.4	-0.85	0.045	mA	1-3
F54	ENABLE CHAIN	84	2.4	-7.15	-0.462	mA	1-3
F55	(C0)	85	2.4	-2.6	0.07	mA	1-3
F56	(C1)	86	2.4	-3.2	0.09	mA	1-3
F57	SYS CLOCK	92	2.4	-2.0	0.05	mA	1-3
F58	SOURCE A 7	97	2.4	-0.72	0.04	mA	1-3
F59	SOURCE A 6	98	|	|	|	|	|
F60	SOURCE A 5	99	|	|	|	|	|
F61	SOURCE A 4	100	2.4	-0.72	0.04	mA	1-3

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NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F61
Power supply current limit	3.06 A	NA	F1-F61
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F61
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	F1-F61
low, (VL)	0.0 V	± 100 mV	F1-F61
Input current measurement	NA	± 38.0 uA	F1-F4
	NA	± 116.0 uA	F5
	NA	± 44.7 uA	F6-F8
	NA	± 102.0 uA	F9
	NA	± 31.0 uA	F10
	NA	± 52.0 uA	F11
	NA	± 13.7 uA	F12
	NA	± 31.0 uA	F13
	NA	± 62.0 uA	F14
	NA	± 13.7 uA	F15,F16
	NA	± 44.7 uA	F17
	NA	± 52.0 uA	F18-F21
	NA	± 44.7 uA	F22
	NA	± 183.0 uA	F23-F25
	NA	± 76.0 uA	F26-F28
	NA	± 38.0 uA	F29-F36
	NA	± 21.0 uA	F37-F39
	NA	± 102.0 uA	F40
	NA	± 205.0 uA	F41
	NA	± 76.0 uA	F42
	NA	± 205.0 uA	F43
	NA	± 63.5 uA	F44

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 42.5 uA	F45
	NA	± 21.0 uA	F46,F47
	NA	± 31.0 uA	F48
	NA	± 205.0 uA	F49
	NA	± 102.0 uA	F50
	NA	± 44.7 uA	F51-F53
	NA	± 330.0 uA	F54
	NA	± 133.0 uA	F55
	NA	± 164.0 uA	F56
	NA	± 102.0 uA	F57
	NA	± 38.0 uA	F58-F61

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	SOURCE A 15	2	0.4	-0.72	0.04	mA	1-3
G2	SOURCE A 14	3					
G3	SOURCE A 13	4					
G4	SOURCE A 12	5	0.4	-0.72	0.04	mA	1-3
G5	BC/MC 03	19	0.4	-2.25	0.075	mA	1-3
G6	BC/MC 02	20	0.4	-0.85	0.045	mA	1-3
G7	BC/MC 01	21					
G8	BC/MC 00	22	0.4	-0.85	0.045	mA	1-3
G9	EX BRAN COND	23	0.4	-2.0	0.05	mA	1-3
G10	MASTER CLEAR	24	0.4	-0.6	0.02	mA	1-3
G11	(SB04)	26	0.4	-1.0	0.04	mA	1-3
G12	MC CL FORMAT	27	0.4	-0.25	0.025	mA	1-3
G13	MC IR (F00)	28	0.4	-0.6	0.02	mA	1-3
G14	TOGGLE ALU	29	0.4	-1.2	0.04	mA	1-3
G15	MCU (00)	30	0.4	-0.25	0.025	mA	1-3
G16	MCU (01)	32	0.4	-0.25	0.025	mA	1-3
G17	(D00)	33	0.4	-0.85	0.045	mA	1-3
G18	(SB03)	35	0.4	-1.0	0.04	mA	1-3
G19	(SB02)	36					
G20	(SB01)	37	0.4	-1.0	0.04	mA	1-3

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Check the source to verify that this is the current version before use.

TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G21	(SB00)	38	0.4	-1.0	0.04	mA	1-3
G22	(D01)	39	0.4	-0.85	0.045	mA	1-3
G23	(F03)	41	0.4	-3.48	0.18	mA	1-3
G24	(F02)	42					
G25	(F01)	43	0.4	-3.48	0.18	mA	1-3
G26	(SA02)	44	0.4	-1.44	0.08	mA	1-3
G27	(SA01)	45					
G28	(SA00)	46	0.4	-1.44	0.08	mA	1-3
G29	SOURCE A 11	47	0.4	-0.72	0.04	mA	1-3
G30	SOURCE A 10	48					
G31	SOURCE A 9	49					
G32	SOURCE A 8	50					
G33	SOURCE A 3	52					
G34	SOURCE A 2	53					
G35	SOURCE A 1	54					
G36	SOURCE A 0	55	0.4	-0.72	0.04	mA	1-3
G37	CLK RATE 2	56	0.4	-0.4	0.02	mA	1-3
G38	CLK RATE 1	57					
G39	CLK RATE 0	58	0.4	-0.4	0.02	mA	1-3
G40	20 MHZ OSC	63	0.4	-2.0	0.05	mA	1-3

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Check the source to verify that this is the current version before use.

TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G41	DESTN CK	64	0.4	-4.0	0.1	mA	1-3
G42	(SA03)	66	0.4	-1.44	0.08	mA	1-3
G43	B BUS ENABLE	67	0.4	-4.0	0.1	mA	1-3
G44	WAIT CLOCK	69	0.4	-1.2	0.07	mA	1-3
G45	WAIT REQUEST	71	0.4	-0.8	0.05	mA	1-3
G46	READY	72	0.4	-0.4	0.02	mA	1-3
G47	SMI DATA OUT	76	0.4	-0.4	0.02	mA	1-3
G48	MICRO IR (S)	77	0.4	-0.6	0.02	mA	1-3
G49	SMI LD/SF CT	78	0.4	-4.0	0.1	mA	1-3
G50	DISABLE ALU	79	0.4	-2.0	0.05	mA	1-3
G51	(D02)	80	0.4	-0.85	0.045	mA	1-3
G52	(D03)	82					
G53	(D04)	83	0.4	-0.85	0.045	mA	1-3
G54	ENABLE CHAIN	84	0.4	-7.15	-0.462	mA	1-3
G55	(C0)	85	0.4	-2.6	0.07	mA	1-3
G56	(C1)	86	0.4	-3.2	0.09	mA	1-3
G57	SYS CLOCK	92	0.4	-2.0	0.05	mA	1-3
G58	SOURCE A 7	97	0.4	-0.72	0.04	mA	1-3
G59	SOURCE A 6	98					
G60	SOURCE A 5	99					
G61	SOURCE A 4	100	0.4	-0.72	0.04	mA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:25Z
Check the source to verify that this is the current version before use.

NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G61
Power supply current limit	3.06 A	NA	G1-G61
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G61
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	G1-G61
low, (VL)	0.0 V	± 100 mV	G1-G61
Input current measurement	NA	± 38.0 uA	G1-G4
	NA	± 116.0 uA	G5
	NA	± 44.7 uA	G6-G8
	NA	± 102.0 uA	G9
	NA	± 31.0 uA	G10
	NA	± 52.0 uA	G11
	NA	± 13.7 uA	G12
	NA	± 31.0 uA	G13
	NA	± 62.0 uA	G14
	NA	± 13.7 uA	G15,G16
	NA	± 44.7 uA	G17
	NA	± 52.0 uA	G18-G21
	NA	± 44.7 uA	G22
	NA	± 183.0 uA	G23-G25
	NA	± 76.0 uA	G26-G28
	NA	± 38.0 uA	G29-G36
	NA	± 21.0 uA	G37-G39
	NA	± 102.0 uA	G40
	NA	± 205.0 uA	G41
	NA	± 76.0 uA	G42
	NA	± 205.0 uA	G43
	NA	± 63.5 uA	G44

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 42.5 uA	G45
	NA	± 21.0 uA	G46,G47
	NA	± 31.0 uA	G48
	NA	± 205.0 uA	G49
	NA	± 102.0 uA	G50
	NA	± 44.7 uA	G51-G53
	NA	± 330.0 uA	G54
	NA	± 133.0 uA	G55
	NA	± 164.0 uA	G56
	NA	± 102.0 uA	G57
	NA	± 38.0 uA	G58-G61

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H1	ALU OVL	6	-0.4	2.4	4.5	V	1-4
H2	ALU CARRY	7					
H3	ALU NEG	8	-0.4	2.4	4.5	V	1-4
H4	DESTN BUS 15	9	-0.16	2.4	4.5	V	1-4
H5	DESTN BUS 15	9					
H6	DESTN BUS 15	9					
H7	DESTN BUS 14	11					
H8	DESTN BUS 14	11					
H9	DESTN BUS 14	11					
H10	DESTN BUS 13	12					
H11	DESTN BUS 13	12					
H12	DESTN BUS 13	12					
H13	DESTN BUS 12	13					
H14	DESTN BUS 12	13					
H15	DESTN BUS 12	13	-0.16	2.4	4.5	V	1-4
H16	JMP TRUE	14	-0.4	2.4	4.5	V	1-4
H17	DESTN BUS 11	15	-0.16	2.4	4.5	V	1-4
H18	DESTN BUS 11	15					
H19	DESTN BUS 11	15					
H20	DESTN BUS 10	16	-0.16	2.4	4.5	V	1-4

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Check the source to verify that this is the current version before use.

TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H21	DESTN BUS 10	16	-0.16	2.4	4.5	V	1-4
H22	DESTN BUS 10	16					
H23	DESTN BUS 9	17					
H24	DESTN BUS 9	17					
H25	DESTN BUS 8	18					
H26	DESTN BUS 8	18					
H27	DESTN BUS 8	18	-0.16	2.4	4.5	V	1-4
H28	ALU ZERO	34	-0.4	2.4	4.5	V	1-4
H29	20 MHZ CLOCK	59	-0.16	2.4	4.5	V	1-4
H30	ALU C1 CLOCK	61	-0.4	2.4	4.5	V	1-4
H31	MP TO MPC	62	-0.16	2.4	4.5	V	1-4
H32	ALU C3 CLOCK	65	-0.4	2.4	4.5	V	1-4
H33	ALU C2 CLOCK	68					
H34	ALU C4 CLOCK	70					
H35	WAIT ACK	73	-0.4	2.4	4.5	V	1-4
H36	SERIAL OUT	74	-0.16	2.4	4.5	V	1-4
H37	DESTN BUS 7	87					
H38	DESTN BUS 7	87					
H39	DESTN BUS 7	87					
H40	DESTN BUS 6	88					
H41	DESTN BUS 6	88	-0.16	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H42	DESTN BUS 6	88	-0.16	2.4	4.5	V	1-4
H43	DESTN BUS 5	89					
H44	DESTN BUS 5	89					
H45	DESTN BUS 5	89					
H46	DESTN BUS 4	91					
H47	DESTN BUS 4	91					
H48	DESTN BUS 4	91					
H49	DESTN BUS 3	93					
H50	DESTN BUS 3	93					
H51	DESTN BUS 3	93					
H52	DESTN BUS 2	94					
H53	DESTN BUS 2	94					
H54	DESTN BUS 2	94					
H55	DESTN BUS 1	95					
H56	DESTN BUS 1	95					
H57	DESTN BUS 1	95					
H58	DESTN BUS 0	96					
H59	DESTN BUS 0	96					
H60	DESTN BUS 0	96	-0.16	2.4	4.5	V	1-4

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NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H60
Power supply current limit	3.06 A	NA	H1-H60
Forcing current, (Ioh)	-0.4 mA	± 40 uA	H1-H3
	-0.16 mA	± 16 uA	H4-H15
	-0.4 mA	± 40 uA	H16
	-0.16 mA	± 16 uA	H17-H27
	-0.4 mA	± 40 uA	H28
	-0.16 mA	± 16 uA	H29
	-0.4 mA	± 40 uA	H30
	-0.16 mA	± 16 uA	H31
	-0.4 mA	± 40 uA	H32-H35
	-0.16 mA	± 16 uA	H36-H60
Pattern voltage high, (VH)	3.0 V	± 100 mV	H1-H60
low, (VL)	0.0 V	± 100 mV	H1-H60
Output voltage measurement	NA	± 105 mV	H1-H60

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

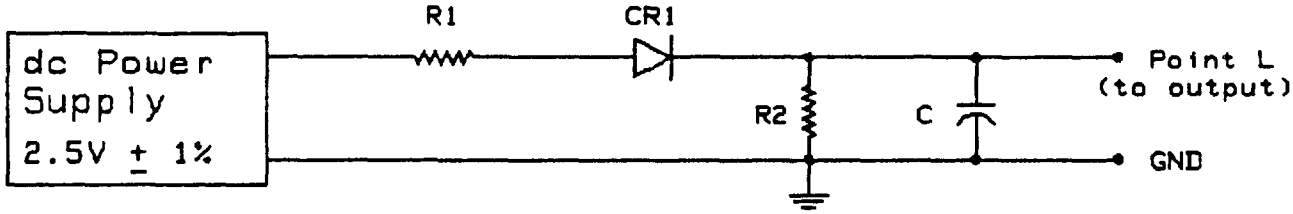
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
6,7,8,14,34	6200	6.0	0.5
61,65,68,70,73	6200	1.6	0.5
9,11,12,13,15,16, 17,18,59,62,74, 87,88,89,91,93, 94,95,96	16000	1.6	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	ALU OVL	6	6.0	0.0	0.5	V	1-4
J2	ALU CARRY	7					
J3	ALU NEG	8	6.0	0.0	0.5	V	1-4
J4	DESTN BUS 15	9	1.6	0.0	0.5	V	1-4
J5	DESTN BUS 15	9					
J6	DESTN BUS 15	9					
J7	DESTN BUS 14	11					
J8	DESTN BUS 14	11					
J9	DESTN BUS 14	11					
J10	DESTN BUS 13	12					
J11	DESTN BUS 13	12					
J12	DESTN BUS 13	12					
J13	DESTN BUS 12	13					
J14	DESTN BUS 12	13					
J15	DESTN BUS 12	13	1.6	0.0	0.5	V	1-4
J16	JMP TRUE	14	6.0	0.0	0.5	V	1-4
J17	DESTN BUS 11	15	1.6	0.0	0.5	V	1-4
J18	DESTN BUS 11	15					
J19	DESTN BUS 11	15					
J20	DESTN BUS 10	16	1.6	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J21	DESTN BUS 10	16	1.6	0.0	0.5	V	1-4
J22	DESTN BUS 10	16					
J23	DESTN BUS 9	17					
J24	DESTN BUS 9	17					
J25	DESTN BUS 9	17					
J26	DESTN BUS 8	18					
J27	DESTN BUS 8	18					
J28	DESTN BUS 8	18	1.6	0.0	0.5	V	1-4
J29	ALU ZERO	34	6.0	0.0	0.5	V	1-4
J30	20 MHZ CLOCK	59	1.6	0.0	0.5	V	1-4
J31	ALU C1 CLOCK	61					
J32	MP TO MPC	62					
J33	ALU C3 CLOCK	65					
J34	ALU C2 CLOCK	68					
J35	ALU C4 CLOCK	70					
J36	WAIT ACK	73					
J37	SERIAL OUT	74					
J38	DESTN BUS 7	87					
J39	DESTN BUS 7	87					
J40	DESTN BUS 7	87	1.6	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J41	DESTN BUS 6	88	1.6	0.0	0.5	V	1-4
J42	DESTN BUS 6	88					
J43	DESTN BUS 6	88					
J44	DESTN BUS 5	89					
J45	DESTN BUS 5	89					
J46	DESTN BUS 5	89					
J47	DESTN BUS 4	91					
J48	DESTN BUS 4	91					
J49	DESTN BUS 4	91					
J50	DESTN BUS 3	93					
J51	DESTN BUS 3	93					
J52	DESTN BUS 3	93					
J53	DESTN BUS 2	94					
J54	DESTN BUS 2	94					
J55	DESTN BUS 2	94					
J56	DESTN BUS 1	95					
J57	DESTN BUS 1	95					
J58	DESTN BUS 1	95					
J59	DESTN BUS 0	96					
J60	DESTN BUS 0	96					
J61	DESTN BUS 0	96	1.6	0.0	0.5	V	1-4

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NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J61
Power supply current limit	3.06 A	NA	J1-J61
Forcing current, (Iol)	6.0 mA 1.6 mA 6.0 mA 1.6 mA 6.0 mA 1.6 mA	± 600 uA ± 160 uA ± 600 uA ± 160 uA ± 600 uA ± 160 uA	J1-J3 J4-J15 J16 J17-J28 J29 J30-J61
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	J1-J61 J1-J61
Output voltage measurement	NA	± 25 mV	J1-J61

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Output leakage current, high and low level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
K1	JMP TRUE	14	2.4	-0.05	0.05	mA	1-3
K2	JMP TRUE	14	0.4	-0.05	0.05	mA	1-3

NOTES:

1. Output leakage current, high and low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	K1,K2
Power supply current limit	3.06 A	NA	K1,K2
Forcing voltage, (Voh) (Vol)	2.4 V 0.4 V	± 50 mV ± 50 mV	K1 K2
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	K1,K2 K1,K2
Output current measurement	NA	± 5 uA	K1,K2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	SOURCE A 15	2	-18.0	-1.5	0.0	V	1-3
L2	SOURCE A 14	3					
L3	SOURCE A 13	4					
L4	SOURCE A 12	5	-18.0	-1.5	0.0	V	1-3
L5	BC/MC 03	19	-18.0	-1.2	0.0	V	1-3
L6	BC/MC 02	20					
L7	BC/MC 01	21					
L8	BC/MC 00	22					
L9	EX BRAN COND	23					
L10	MASTER CLEAR	24					
L11	(SB04)	26	-18.0	-1.2	0.0	V	1-3
L12	MC CL FORMAT	27	-18.0	-1.5	0.0	V	1-3
L13	MC IR (F00)	28	-18.0	-1.2	0.0	V	1-3
L14	TOGGLE ALU	29	-18.0	-1.2	0.0	V	1-3
L15	MCU (00)	30	-18.0	-1.5	0.0	V	1-3
L16	MCU (01)	32	-18.0	-1.5	0.0	V	1-3
L17	(D00)	33	-18.0	-1.2	0.0	V	1-3
L18	(SB03)	35					
L19	(SB02)	36					
L20	(SB01)	37	-18.0	-1.2	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
L21	(SB00)	38	-18.0	-1.2	0.0	V	1-3
L22	(D01)	39					
L23	(F03)	41					
L24	(F02)	42					
L25	(F01)	43	-18.0	-1.2	0.0	V	1-3
L26	(SA02)	44	-18.0	-1.5	0.0	V	1-3
L27	(SA01)	45					
L28	(SA00)	46					
L29	SOURCE A 11	47					
L30	SOURCE A 10	48					
L31	SOURCE A 9	49					
L32	SOURCE A 8	50					
L33	SOURCE A 3	52					
L34	SOURCE A 2	53					
L35	SOURCE A 1	54					
L36	SOURCE A 0	55					
L37	CLK RATE 2	56					
L38	CLK RATE 1	57					
L39	CLK RATE 0	58	-18.0	-1.5	0.0	V	1-3
L40	20 MHZ OSC	63	-18.0	-1.2	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L41	DESTN CK	64	-18.0	-1.2	0.0	V	1-3
L42	(SA03)	66	-18.0	-1.5	0.0	V	1-3
L43	B BUS ENABLE	67	-18.0	-1.2	0.0	V	1-3
L44	WAIT CLOCK	69	-18.0	-1.5	0.0	V	1-3
L45	WAIT REQUEST	71					
L46	READY	72					
L47	SMI DAT OUT	76	-18.0	-1.5	0.0	V	1-3
L48	MICRO IR (S)	77	-18.0	-1.2	0.0	V	1-3
L49	SMI LD/SF CT	78					
L50	DISABLE ALU	79					
L51	(D02)	80					
L52	(D03)	82					
L53	(D04)	83					
L54	(C0)	85					
L55	(C1)	86					
L56	SYS CLOCK	92	-18.0	-1.2	0.0	V	1-3
L57	SOURCE A 7	97	-18.0	-1.5	0.0	V	1-3
L58	SOURCE A 6	98					
L59	SOURCE A 5	99					
L60	SOURCE A 4	100	-18.0	-1.5	0.0	V	1-3

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NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	L1-L60
Power supply current limit	NA	NA	L1-L60
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L60
Pattern voltage high, (VH) low, (VL)	 NA NA	 NA NA	 L1-L60 L1-L60
Input voltage measurement	NA NA NA NA NA NA NA NA NA NA NA NA NA	± 75 mV ± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV ± 75 mV	L1-L4 L5-L11 L12 L13,L14 L15,L16 L17-L25 L26-L39 L40,L41 L42 L43 L44-L47 L48-L56 L57-L60

2. The following procedure is required to perform this test:

   (a) No power shall be applied to the module.

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NOTES: - Continued.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XII and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M1	PSFC	1	3.8	7.0	uF	1-4
M2	PSFC	51	5.9	11.0	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M1,M2
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M1,M2
Capacitance measurement	NA NA	± 160 nF ± 225 nF	M1 M2

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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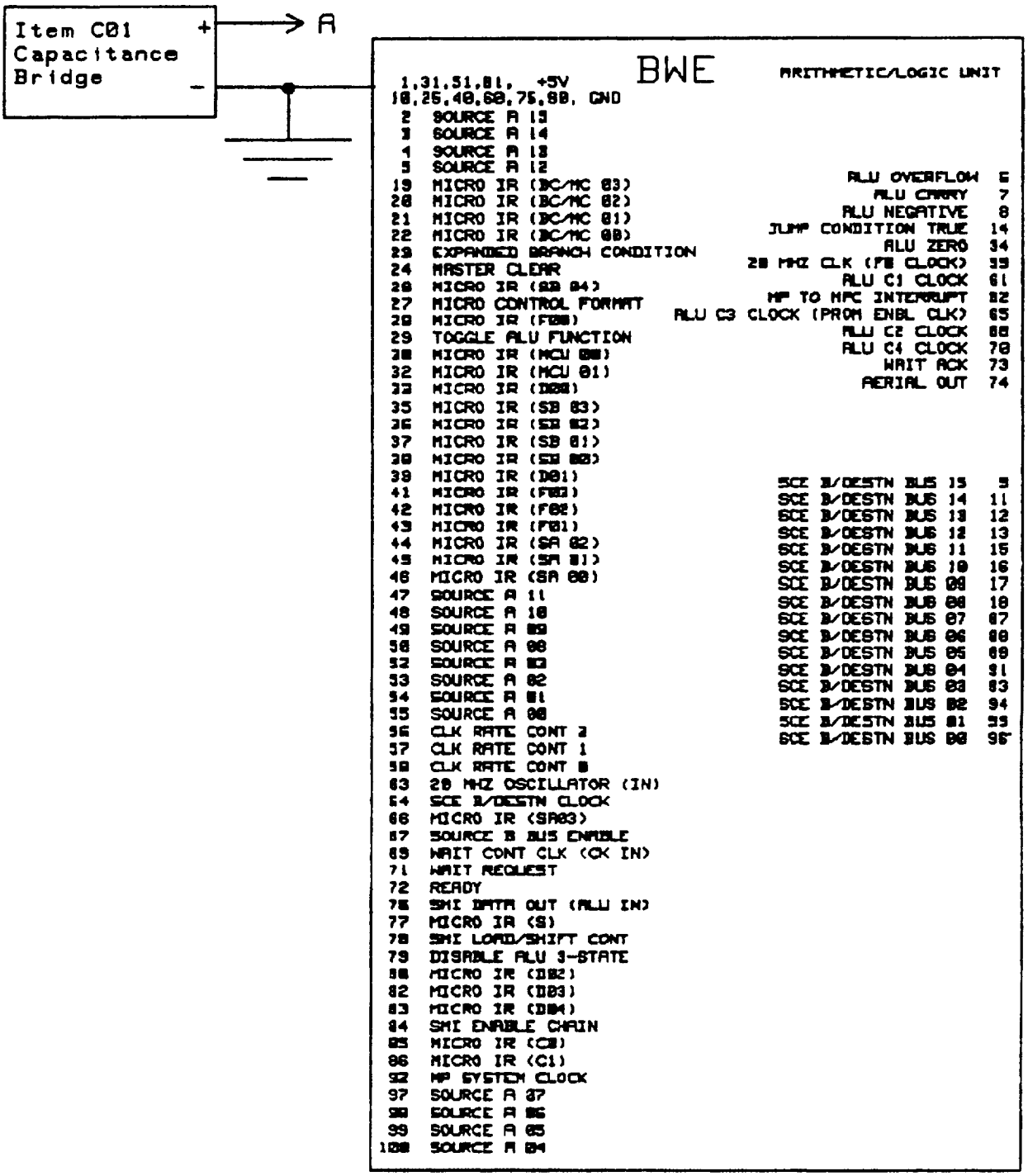


FIGURE 10. Power supply filter capacitance test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1,31, 51,81	850	2040	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	3.06 A	NA	N
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 N N
Power supply current measurement	NA	± 65 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes	Figs
P1	Functional test minimum power supply voltage maximum pattern rate (300 nS)	1-3	9,11
P2	Functional test maximum power supply voltage maximum pattern rate (300 nS)	1-3	9,11

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	3.06 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 25 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

(c) The module inputs shall not violate the minimum input timing parameters reference on figure 11.

3. Abbreviations used are defined in 6.1.

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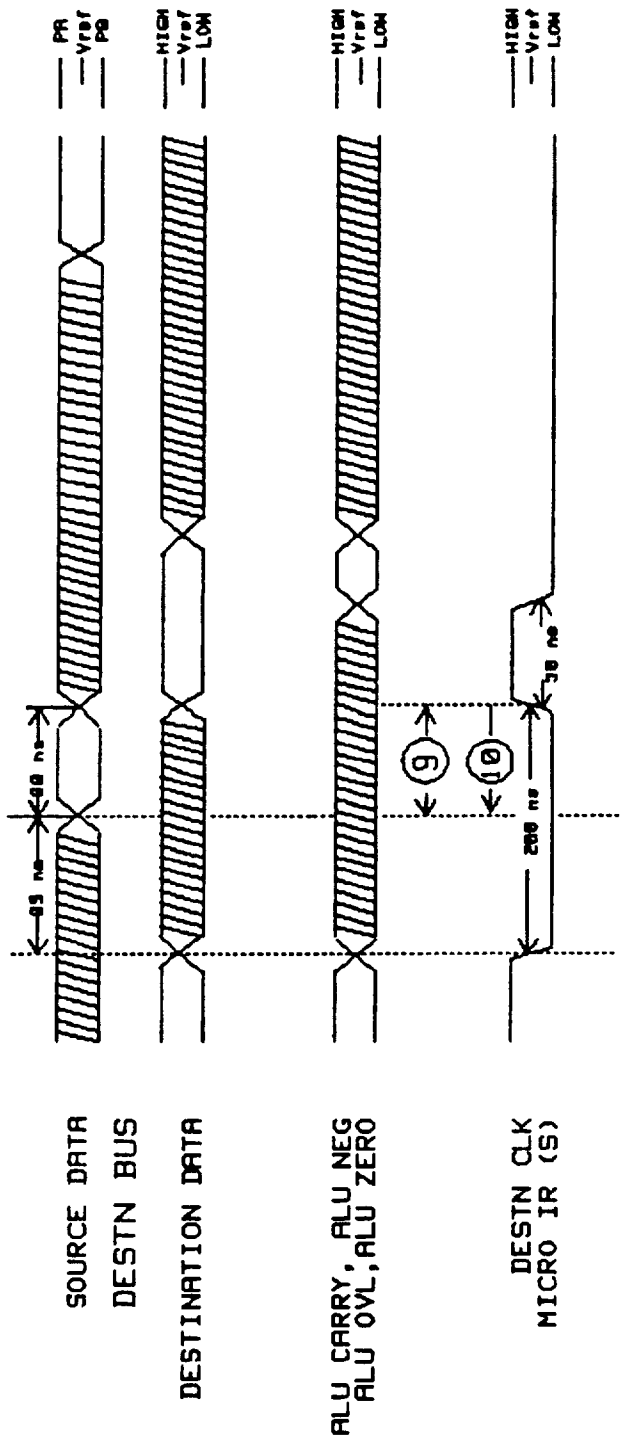


FIGURE 11. Input timing parameters for table XV.

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NOTES:

1. (9) DESTN BUS BITS to DESTN CK. Setup time: minimum of 165 nS.

2. (10) DESTN BUS BITS to MICRO IR (S). Setup time: minimum of 165 nS.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (300 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (300 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (300 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (300 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	3.06 A	NA	Q1-Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 100 mV	Q1,Q3
high, (Voh)	NA	± 100 mV	Q2,Q4
low, (Vol)	NA	± 100 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

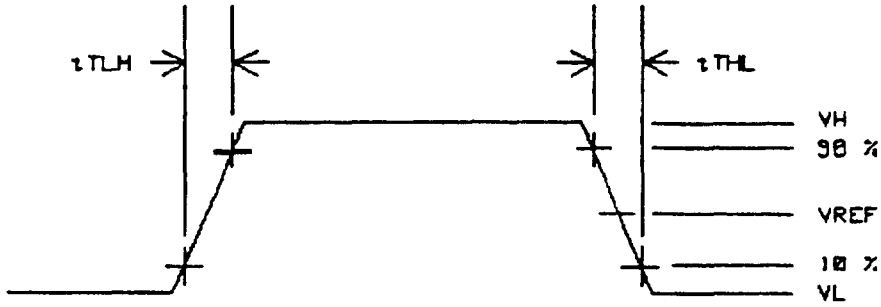
(b) The output compare levels shall be as follows:

High level: Vref ≤ Voh ≤ Vcc
Low level: 0.0 V ≤ Vol ≤ Vref

(c) The dynamic parameters are found in table XVII.

3. The load circuit for this test is specified on figure 9.

4. Input/output waveforms are as referenced on figure 12. All of the input waveforms shall have the following characteristics:



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NOTES: - Continued.

Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XVI.

6. Abbreviations used are defined in 6.1.

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TABLE XVII. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	DESTN CK to DESTN BUS 15	-	-	-	85	85	85	nS	12
1	tpd	DESTN CK to DESTN BUS 14	-	-	-	85	85	85	nS	12
1	tpd	DESTN CK to DESTN BUS 13	-	-	-	85	85	85	nS	12
1	tpd	DESTN CK to DESTN BUS 12	-	-	-	85	85	85	nS	12
1	tpd	DESTN CK to DESTN BUS 11	-	-	-	85	85	85	nS	12
1	tpd	DESTN CK to DESTN BUS 10	-	-	-	85	85	85	nS	12
1	tpd	DESTN CK to DESTN BUS 9	-	-	-	85	85	85	nS	12
1	tpd	DESTN CK to DESTN BUS 8	-	-	-	85	85	85	nS	12
1	tpd	DESTN CK to DESTN BUS 7	-	-	-	85	85	85	nS	12
1	tpd	DESTN CK to DESTN BUS 6	-	-	-	85	85	85	nS	12
1	tpd	DESTN CK to DESTN BUS 5	-	-	-	85	85	85	nS	12
1	tpd	DESTN CK to DESTN BUS 4	-	-	-	85	85	85	nS	12
1	tpd	DESTN CK to DESTN BUS 3	-	-	-	85	85	85	nS	12
1	tpd	DESTN CK to DESTN BUS 2	-	-	-	85	85	85	nS	12
1	tpd	DESTN CK to DESTN BUS 1	-	-	-	85	85	85	nS	12
1	tpd	DESTN CK to DESTN BUS 0	-	-	-	85	85	85	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tpd	(SB04) to DESTN BUS 15	-	-	-	165	165	165	nS	12
2	tpd	(SB04) to DESTN BUS 14	-	-	-	165	165	165	nS	12
2	tpd	(SB04) to DESTN BUS 13	-	-	-	165	165	165	nS	12
2	tpd	(SB04) to DESTN BUS 12	-	-	-	165	165	165	nS	12
2	tpd	(SB04) to DESTN BUS 11	-	-	-	165	165	165	nS	12
2	tpd	(SB04) to DESTN BUS 10	-	-	-	165	165	165	nS	12
2	tpd	(SB04) to DESTN BUS 9	-	-	-	165	165	165	nS	12
2	tpd	(SB04) to DESTN BUS 8	-	-	-	165	165	165	nS	12
2	tpd	(SB04) to DESTN BUS 7	-	-	-	165	165	165	nS	12
2	tpd	(SB04) to DESTN BUS 6	-	-	-	165	165	165	nS	12
2	tpd	(SB04) to DESTN BUS 5	-	-	-	165	165	165	nS	12
2	tpd	(SB04) to DESTN BUS 4	-	-	-	165	165	165	nS	12
		to DESTN BUS 3								

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tpd	(SB04) to DESTN BUS 2	-	-	-	165	165	165	nS	12
2	tpd	(SB04) to DESTN BUS 1	-	-	-	165	165	165	nS	12
2	tpd	(SB04) to DESTN BUS 0	-	-	-	165	165	165	nS	12
2	tpd	MC IR (F00) to DESTN BUS 15	-	-	-	165	165	165	nS	12
2	tpd	MC IR (F00) to DESTN BUS 14	-	-	-	165	165	165	nS	12
2	tpd	MC IR (F00) to DESTN BUS 13	-	-	-	165	165	165	nS	12
2	tpd	MC IR (F00) to DESTN BUS 12	-	-	-	165	165	165	nS	12
2	tpd	MC IR (F00) to DESTN BUS 11	-	-	-	165	165	165	nS	12
2	tpd	MC IR (F00) to DESTN BUS 10	-	-	-	165	165	165	nS	12
2	tpd	MC IR (F00) to DESTN BUS 9	-	-	-	165	165	165	nS	12
2	tpd	MC IR (F00) to DESTN BUS 8	-	-	-	165	165	165	nS	12
2	tpd	MC IR (F00) to DESTN BUS 7	-	-	-	165	165	165	nS	12
2	tpd	MC IR (F00) to DESTN BUS 6	-	-	-	165	165	165	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tpd	MC IR (F00) to DESTN BUS 5	-	-	-	165	165	165	nS	12
2	tpd	MC IR (F00) to DESTN BUS 4	-	-	-	165	165	165	nS	12
2	tpd	MC IR (F00) to DESTN BUS 3	-	-	-	165	165	165	nS	12
2	tpd	MC IR (F00) to DESTN BUS 2	-	-	-	165	165	165	nS	12
2	tpd	MC IR (F00) to DESTN BUS 1	-	-	-	165	165	165	nS	12
2	tpd	MC IR (F00) to DESTN BUS 0	-	-	-	165	165	165	nS	12
2	tpd	(SB03) to DESTN BUS 15	-	-	-	165	165	165	nS	12
2	tpd	(SB03) to DESTN BUS 14	-	-	-	165	165	165	nS	12
2	tpd	(SB03) to DESTN BUS 13	-	-	-	165	165	165	nS	12
2	tpd	(SB03) to DESTN BUS 12	-	-	-	165	165	165	nS	12
2	tpd	(SB03) to DESTN BUS 11	-	-	-	165	165	165	nS	12
2	tpd	(SB03) to DESTN BUS 10	-	-	-	165	165	165	nS	12
2	tpd	(SB03) to DESTN BUS 9	-	-	-	165	165	165	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tpd	(SB03) to DESTN BUS 8	-	-	-	165	165	165	nS	12
2	tpd	(SB03) to DESTN BUS 7	-	-	-	165	165	165	nS	12
2	tpd	(SB03) to DESTN BUS 6	-	-	-	165	165	165	nS	12
2	tpd	(SB03) to DESTN BUS 5	-	-	-	165	165	165	nS	12
2	tpd	(SB03) to DESTN BUS 4	-	-	-	165	165	165	nS	12
2	tpd	(SB03) to DESTN BUS 3	-	-	-	165	165	165	nS	12
2	tpd	(SB03) to DESTN BUS 2	-	-	-	165	165	165	nS	12
2	tpd	(SB03) to DESTN BUS 1	-	-	-	165	165	165	nS	12
2	tpd	(SB03) to DESTN BUS 0	-	-	-	165	165	165	nS	12
2	tpd	(SB02) to DESTN BUS 15	-	-	-	165	165	165	nS	12
2	tpd	(SB02) to DESTN BUS 14	-	-	-	165	165	165	nS	12
2	tpd	(SB02) to DESTN BUS 13	-	-	-	165	165	165	nS	12
2	tpd	(SB02) to DESTN BUS 12	-	-	-	165	165	165	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tpd	(SB02) to DESTN BUS 11	-	-	-	165	165	165	nS	12
2	tpd	(SB02) to DESTN BUS 10	-	-	-	165	165	165	nS	12
2	tpd	(SB02) to DESTN BUS 9	-	-	-	165	165	165	nS	12
2	tpd	(SB02) to DESTN BUS 8	-	-	-	165	165	165	nS	12
2	tpd	(SB02) to DESTN BUS 7	-	-	-	165	165	165	nS	12
2	tpd	(SB02) to DESTN BUS 6	-	-	-	165	165	165	nS	12
2	tpd	(SB02) to DESTN BUS 5	-	-	-	165	165	165	nS	12
2	tpd	(SB02) to DESTN BUS 4	-	-	-	165	165	165	nS	12
2	tpd	(SB02) to DESTN BUS 3	-	-	-	165	165	165	nS	12
2	tpd	(SB02) to DESTN BUS 2	-	-	-	165	165	165	nS	12
2	tpd	(SB02) to DESTN BUS 1	-	-	-	165	165	165	nS	12
2	tpd	(SB02) to DESTN BUS 0	-	-	-	165	165	165	nS	12
2	tpd	(SB01) to DESTN BUS 15	-	-	-	165	165	165	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tpd	(SB01) to DESTN BUS 14	-	-	-	165	165	165	nS	12
2	tpd	(SB01) to DESTN BUS 13	-	-	-	165	165	165	nS	12
2	tpd	(SB01) to DESTN BUS 12	-	-	-	165	165	165	nS	12
2	tpd	(SB01) to DESTN BUS 11	-	-	-	165	165	165	nS	12
2	tpd	(SB01) to DESTN BUS 10	-	-	-	165	165	165	nS	12
2	tpd	(SB01) to DESTN BUS 9	-	-	-	165	165	165	nS	12
2	tpd	(SB01) to DESTN BUS 8	-	-	-	165	165	165	nS	12
2	tpd	(SB01) to DESTN BUS 7	-	-	-	165	165	165	nS	12
2	tpd	(SB01) to DESTN BUS 6	-	-	-	165	165	165	nS	12
2	tpd	(SB01) to DESTN BUS 5	-	-	-	165	165	165	nS	12
2	tpd	(SB01) to DESTN BUS 4	-	-	-	165	165	165	nS	12
2	tpd	(SB01) to DESTN BUS 3	-	-	-	165	165	165	nS	12
2	tpd	(SB01) to DESTN BUS 2	-	-	-	165	165	165	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tpd	(SB01) to DESTN BUS 1	-	-	-	165	165	165	nS	12
2	tpd	(SB01) to DESTN BUS 0	-	-	-	165	165	165	nS	12
2	tpd	(SB00) to DESTN BUS 15	-	-	-	165	165	165	nS	12
2	tpd	(SB00) to DESTN BUS 14	-	-	-	165	165	165	nS	12
2	tpd	(SB00) to DESTN BUS 13	-	-	-	165	165	165	nS	12
2	tpd	(SB00) to DESTN BUS 12	-	-	-	165	165	165	nS	12
2	tpd	(SB00) to DESTN BUS 11	-	-	-	165	165	165	nS	12
2	tpd	(SB00) to DESTN BUS 10	-	-	-	165	165	165	nS	12
2	tpd	(SB00) to DESTN BUS 9	-	-	-	165	165	165	nS	12
2	tpd	(SB00) to DESTN BUS 8	-	-	-	165	165	165	nS	12
2	tpd	(SB00) to DESTN BUS 7	-	-	-	165	165	165	nS	12
2	tpd	(SB00) to DESTN BUS 6	-	-	-	165	165	165	nS	12
		to DESTN BUS 5								

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tpd	(SB00) to DESTN BUS 4	-	-	-	165	165	165	nS	12
2	tpd	(SB00) to DESTN BUS 3	-	-	-	165	165	165	nS	12
2	tpd	(SB00) to DESTN BUS 2	-	-	-	165	165	165	nS	12
2	tpd	(SB00) to DESTN BUS 1	-	-	-	165	165	165	nS	12
2	tpd	(SB00) to DESTN BUS 0	-	-	-	165	165	165	nS	12
2	tpd	(F03) to DESTN BUS 15	-	-	-	165	165	165	nS	12
2	tpd	(F03) to DESTN BUS 14	-	-	-	165	165	165	nS	12
2	tpd	(F03) to DESTN BUS 13	-	-	-	165	165	165	nS	12
2	tpd	(F03) to DESTN BUS 12	-	-	-	165	165	165	nS	12
2	tpd	(F03) to DESTN BUS 11	-	-	-	165	165	165	nS	12
2	tpd	(F03) to DESTN BUS 10	-	-	-	165	165	165	nS	12
2	tpd	(F03) to DESTN BUS 9	-	-	-	165	165	165	nS	12
2	tpd	(F03) to DESTN BUS 8	-	-	-	165	165	165	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tpd	(F03) to DESTN BUS 7	-	-	-	165	165	165	nS	12
2	tpd	(F03) to DESTN BUS 6	-	-	-	165	165	165	nS	12
2	tpd	(F03) to DESTN BUS 5	-	-	-	165	165	165	nS	12
2	tpd	(F03) to DESTN BUS 4	-	-	-	165	165	165	nS	12
2	tpd	(F03) to DESTN BUS 3	-	-	-	165	165	165	nS	12
2	tpd	(F03) to DESTN BUS 2	-	-	-	165	165	165	nS	12
2	tpd	(F03) to DESTN BUS 1	-	-	-	165	165	165	nS	12
2	tpd	(F03) to DESTN BUS 0	-	-	-	165	165	165	nS	12
2	tpd	(F02) to DESTN BUS 15	-	-	-	165	165	165	nS	12
2	tpd	(F02) to DESTN BUS 14	-	-	-	165	165	165	nS	12
2	tpd	(F02) to DESTN BUS 13	-	-	-	165	165	165	nS	12
2	tpd	(F02) to DESTN BUS 12	-	-	-	165	165	165	nS	12
2	tpd	(F02) to DESTN BUS 11	-	-	-	165	165	165	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tpd	(F02) to DESTN BUS 10	-	-	-	165	165	165	nS	12
2	tpd	(F02) to DESTN BUS 9	-	-	-	165	165	165	nS	12
2	tpd	(F02) to DESTN BUS 8	-	-	-	165	165	165	nS	12
2	tpd	(F02) to DESTN BUS 7	-	-	-	165	165	165	nS	12
2	tpd	(F02) to DESTN BUS 6	-	-	-	165	165	165	nS	12
2	tpd	(F02) to DESTN BUS 5	-	-	-	165	165	165	nS	12
2	tpd	(F02) to DESTN BUS 4	-	-	-	165	165	165	nS	12
2	tpd	(F02) to DESTN BUS 3	-	-	-	165	165	165	nS	12
2	tpd	(F02) to DESTN BUS 2	-	-	-	165	165	165	nS	12
2	tpd	(F02) to DESTN BUS 1	-	-	-	165	165	165	nS	12
2	tpd	(F02) to DESTN BUS 0	-	-	-	165	165	165	nS	12
2	tpd	(F01) to DESTN BUS 15	-	-	-	165	165	165	nS	12
2	tpd	(F01) to DESTN BUS 14	-	-	-	165	165	165	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tpd	(F01) to DESTN BUS 13	-	-	-	165	165	165	nS	12
2	tpd	(F01) to DESTN BUS 12	-	-	-	165	165	165	nS	12
2	tpd	(F01) to DESTN BUS 11	-	-	-	165	165	165	nS	12
2	tpd	(F01) to DESTN BUS 10	-	-	-	165	165	165	nS	12
2	tpd	(F01) to DESTN BUS 9	-	-	-	165	165	165	nS	12
2	tpd	(F01) to DESTN BUS 8	-	-	-	165	165	165	nS	12
2	tpd	(F01) to DESTN BUS 7	-	-	-	165	165	165	nS	12
2	tpd	(F01) to DESTN BUS 6	-	-	-	165	165	165	nS	12
2	tpd	(F01) to DESTN BUS 5	-	-	-	165	165	165	nS	12
2	tpd	(F01) to DESTN BUS 4	-	-	-	165	165	165	nS	12
2	tpd	(F01) to DESTN BUS 3	-	-	-	165	165	165	nS	12
2	tpd	(F01) to DESTN BUS 2	-	-	-	165	165	165	nS	12
2	tpd	(F01) to DESTN BUS 1	-	-	-	165	165	165	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tpd	(F01) to DESTN BUS 0	-	-	-	165	165	165	nS	12
2	tpd	(SA03) to DESTN BUS 15	-	-	-	165	165	165	nS	12
2	tpd	(SA03) to DESTN BUS 14	-	-	-	165	165	165	nS	12
2	tpd	(SA03) to DESTN BUS 13	-	-	-	165	165	165	nS	12
2	tpd	(SA03) to DESTN BUS 12	-	-	-	165	165	165	nS	12
2	tpd	(SA03) to DESTN BUS 11	-	-	-	165	165	165	nS	12
2	tpd	(SA03) to DESTN BUS 10	-	-	-	165	165	165	nS	12
2	tpd	(SA03) to DESTN BUS 9	-	-	-	165	165	165	nS	12
2	tpd	(SA03) to DESTN BUS 8	-	-	-	165	165	165	nS	12
2	tpd	(SA03) to DESTN BUS 7	-	-	-	165	165	165	nS	12
2	tpd	(SA03) to DESTN BUS 6	-	-	-	165	165	165	nS	12
2	tpd	(SA03) to DESTN BUS 5	-	-	-	165	165	165	nS	12
2	tpd	(SA03) to DESTN BUS 4	-	-	-	165	165	165	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tpd	(SA03) to DESTN BUS 3	-	-	-	165	165	165	nS	12
2	tpd	(SA03) to DESTN BUS 2	-	-	-	165	165	165	nS	12
2	tpd	(SA03) to DESTN BUS 1	-	-	-	165	165	165	nS	12
2	tpd	(SA03) to DESTN BUS 0	-	-	-	165	165	165	nS	12
3	tpd	SOURCE A 15 to DESTN BUS 15	-	-	-	95	95	95	nS	12
3	tpd	SOURCE A 14 to DESTN BUS 14	-	-	-	95	95	95	nS	12
3	tpd	SOURCE A 13 to DESTN BUS 13	-	-	-	95	95	95	nS	12
3	tpd	SOURCE A 12 to DESTN BUS 12	-	-	-	95	95	95	nS	12
3	tpd	SOURCE A 11 to DESTN BUS 11	-	-	-	95	95	95	nS	12
3	tpd	SOURCE A 10 to DESTN BUS 10	-	-	-	95	95	95	nS	12
3	tpd	SOURCE A 9 to DESTN BUS 9	-	-	-	95	95	95	nS	12
3	tpd	SOURCE A 8 to DESTN BUS 8	-	-	-	95	95	95	nS	12
3	tpd	SOURCE A 7 to DESTN BUS 7	-	-	-	95	95	95	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
3	tpd	SOURCE A 6 to DESTN BUS 6	-	-	-	95	95	95	nS	12
3	tpd	SOURCE A 5 to DESTN BUS 5	-	-	-	95	95	95	nS	12
3	tpd	SOURCE A 4 to DESTN BUS 4	-	-	-	95	95	95	nS	12
3	tpd	SOURCE A 3 to DESTN BUS 3	-	-	-	95	95	95	nS	12
3	tpd	SOURCE A 2 to DESTN BUS 2	-	-	-	95	95	95	nS	12
3	tpd	SOURCE A 1 to DESTN BUS 1	-	-	-	95	95	95	nS	12
3	tpd	SOURCE A 0 to DESTN BUS 0	-	-	-	95	95	95	nS	12
4	tpd	TOGGLE ALU to DESTN BUS 15	-	-	-	105	105	105	nS	12
4	tpd	TOGGLE ALU to DESTN BUS 14	-	-	-	105	105	105	nS	12
4	tpd	TOGGLE ALU to DESTN BUS 13	-	-	-	105	105	105	nS	12
4	tpd	TOGGLE ALU to DESTN BUS 12	-	-	-	105	105	105	nS	12
4	tpd	TOGGLE ALU to DESTN BUS 11	-	-	-	105	105	105	nS	12
4	tpd	TOGGLE ALU to DESTN BUS 10	-	-	-	105	105	105	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tpd	TOGGLE ALU to DESTN BUS 9	-	-	-	105	105	105	nS	12
4	tpd	TOGGLE ALU to DESTN BUS 8	-	-	-	105	105	105	nS	12
4	tpd	TOGGLE ALU to DESTN BUS 7	-	-	-	105	105	105	nS	12
4	tpd	TOGGLE ALU to DESTN BUS 6	-	-	-	105	105	105	nS	12
4	tpd	TOGGLE ALU to DESTN BUS 5	-	-	-	105	105	105	nS	12
4	tpd	TOGGLE ALU to DESTN BUS 4	-	-	-	105	105	105	nS	12
4	tpd	TOGGLE ALU to DESTN BUS 3	-	-	-	105	105	105	nS	12
4	tpd	TOGGLE ALU to DESTN BUS 2	-	-	-	105	105	105	nS	12
4	tpd	TOGGLE ALU to DESTN BUS 1	-	-	-	105	105	105	nS	12
4	tpd	TOGGLE ALU to DESTN BUS 0	-	-	-	105	105	105	nS	12
5	tpd	DESTN BUS 15 to ALU ZERO	-	-	-	150	150	150	nS	12
5	tpd	DESTN BUS 14 to ALU ZERO	-	-	-	150	150	150	nS	12
5	tpd	DESTN BUS 13 to ALU ZERO	-	-	-	150	150	150	nS	12
5	tpd	DESTN BUS 12 to ALU ZERO	-	-	-	150	150	150	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
5	tpd	DESTN BUS 11 to ALU ZERO	-	-	-	150	150	150	nS	12
5	tpd	DESTN BUS 10 to ALU ZERO	-	-	-	150	150	150	nS	12
5	tpd	DESTN BUS 9 to ALU ZERO	-	-	-	150	150	150	nS	12
5	tpd	DESTN BUS 8 to ALU ZERO	-	-	-	150	150	150	nS	12
5	tpd	DESTN BUS 7 to ALU ZERO	-	-	-	150	150	150	nS	12
5	tpd	DESTN BUS 6 to ALU ZERO	-	-	-	150	150	150	nS	12
5	tpd	DESTN BUS 5 to ALU ZERO	-	-	-	150	150	150	nS	12
5	tpd	DESTN BUS 4 to ALU ZERO	-	-	-	150	150	150	nS	12
5	tpd	DESTN BUS 3 to ALU ZERO	-	-	-	150	150	150	nS	12
5	tpd	DESTN BUS 2 to ALU ZERO	-	-	-	150	150	150	nS	12
5	tpd	DESTN BUS 1 to ALU ZERO	-	-	-	150	150	150	nS	12
5	tpd	DESTN BUS 0 to ALU ZERO	-	-	-	150	150	150	nS	12
6	tpd	MICRO IR (S) to JMP TRUE	-	-	-	65	65	65	nS	12
7	tpd	BC/MC 03 to JMP TRUE	-	-	-	40	40	40	nS	12
7	tpd	BC/MC 00 to JMP TRUE	-	-	-	40	40	40	nS	12
8	tpd	SYS CLOCK to SERIAL OUT	-	-	-	60	60	60	nS	12
9	tsu	DESTN BUS 15 to DESTN CK	165	165	165	-	-	-	nS	12
9	tsu	DESTN BUS 14 to DESTN CK	165	165	165	-	-	-	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
9	tsu	DESTN BUS 13 to DESTN CK	165	165	165	-	-	-	nS	12
9	tsu	DESTN BUS 12 to DESTN CK	165	165	165	-	-	-	nS	12
9	tsu	DESTN BUS 11 to DESTN CK	165	165	165	-	-	-	nS	12
9	tsu	DESTN BUS 10 to DESTN CK	165	165	165	-	-	-	nS	12
9	tsu	DESTN BUS 9 to DESTN CK	165	165	165	-	-	-	nS	12
9	tsu	DESTN BUS 8 to DESTN CK	165	165	165	-	-	-	nS	12
9	tsu	DESTN BUS 7 to DESTN CK	165	165	165	-	-	-	nS	12
9	tsu	DESTN BUS 6 to DESTN CK	165	165	165	-	-	-	nS	12
9	tsu	DESTN BUS 5 to DESTN CK	165	165	165	-	-	-	nS	12
9	tsu	DESTN BUS 4 to DESTN CK	165	165	165	-	-	-	nS	12
9	tsu	DESTN BUS 3 to DESTN CK	165	165	165	-	-	-	nS	12
9	tsu	DESTN BUS 2 to DESTN CK	165	165	165	-	-	-	nS	12
9	tsu	DESTN BUS 1 to DESTN CK	165	165	165	-	-	-	nS	12
9	tsu	DESTN BUS 0 to DESTN CK	165	165	165	-	-	-	nS	12
10	tsu	DESTN BUS 15 to MICRO IR (S)	165	165	165	-	-	-	nS	12
10	tsu	DESTN BUS 14 to MICRO IR (S)	165	165	165	-	-	-	nS	12
10	tsu	DESTN BUS 13 to MICRO IR (S)	165	165	165	-	-	-	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
10	tsu	DESTN BUS 12 to MICRO IR (S)	165	165	165	-	-	-	nS	12
10	tsu	DESTN BUS 11 to MICRO IR (S)	165	165	165	-	-	-	nS	12
10	tsu	DESTN BUS 10 to MICRO IR (S)	165	165	165	-	-	-	nS	12
10	tsu	DESTN BUS 9 to MICRO IR (S)	165	165	165	-	-	-	nS	12
10	tsu	DESTN BUS 8 to MICRO IR (S)	165	165	165	-	-	-	nS	12
10	tsu	DESTN BUS 7 to MICRO IR (S)	165	165	165	-	-	-	nS	12
10	tsu	DESTN BUS 6 to MICRO IR (S)	165	165	165	-	-	-	nS	12
10	tsu	DESTN BUS 5 to MICRO IR (S)	165	165	165	-	-	-	nS	12
10	tsu	DESTN BUS 4 to MICRO IR (S)	165	165	165	-	-	-	nS	12
10	tsu	DESTN BUS 3 to MICRO IR (S)	165	165	165	-	-	-	nS	12
10	tsu	DESTN BUS 2 to MICRO IR (S)	165	165	165	-	-	-	nS	12
10	tsu	DESTN BUS 1 to MICRO IR (S)	165	165	165	-	-	-	nS	12
10	tsu	DESTN BUS 0 to MICRO IR (S)	165	165	165	-	-	-	nS	12

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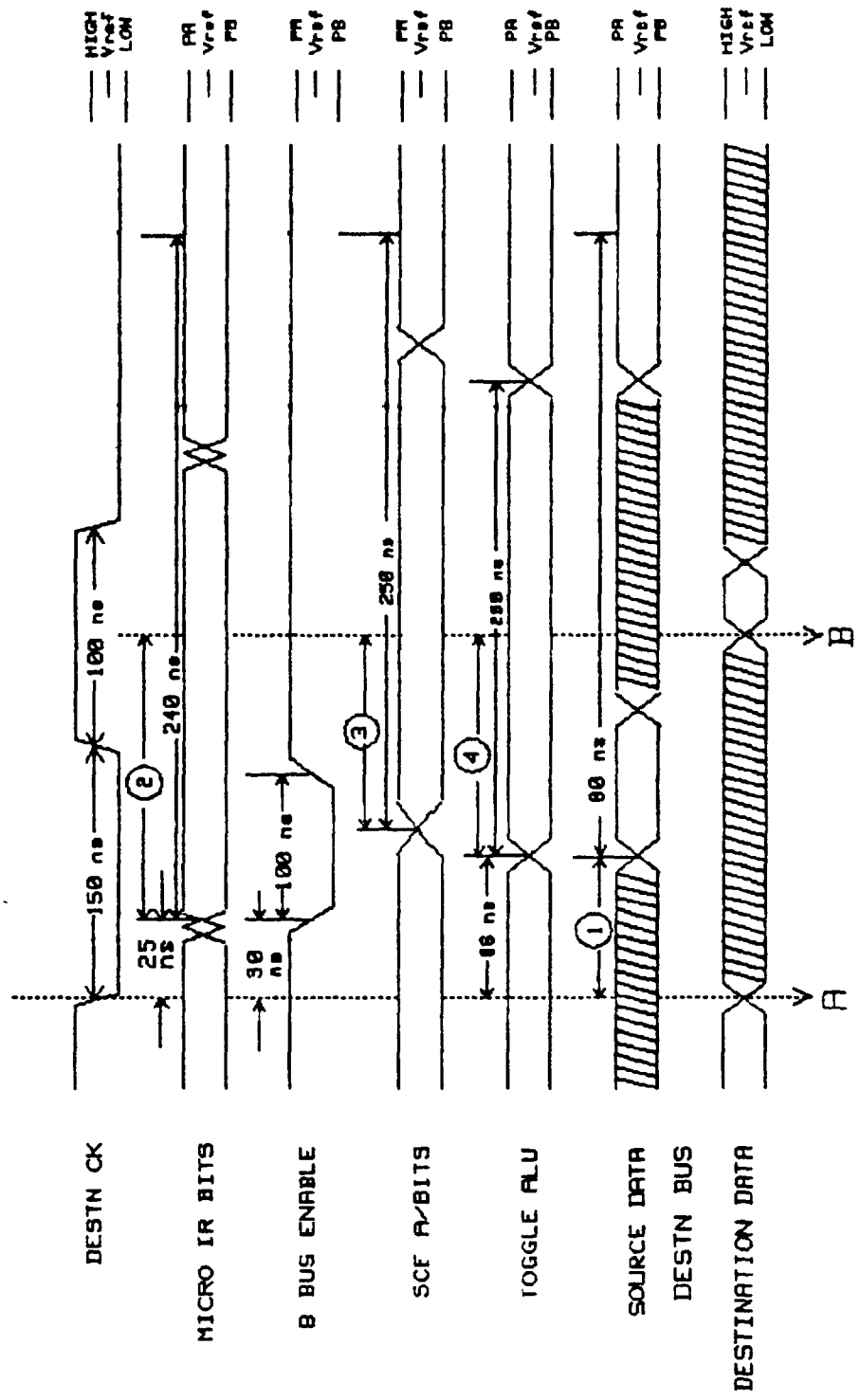


FIGURE 12a. Dynamic waveforms for table XVII.

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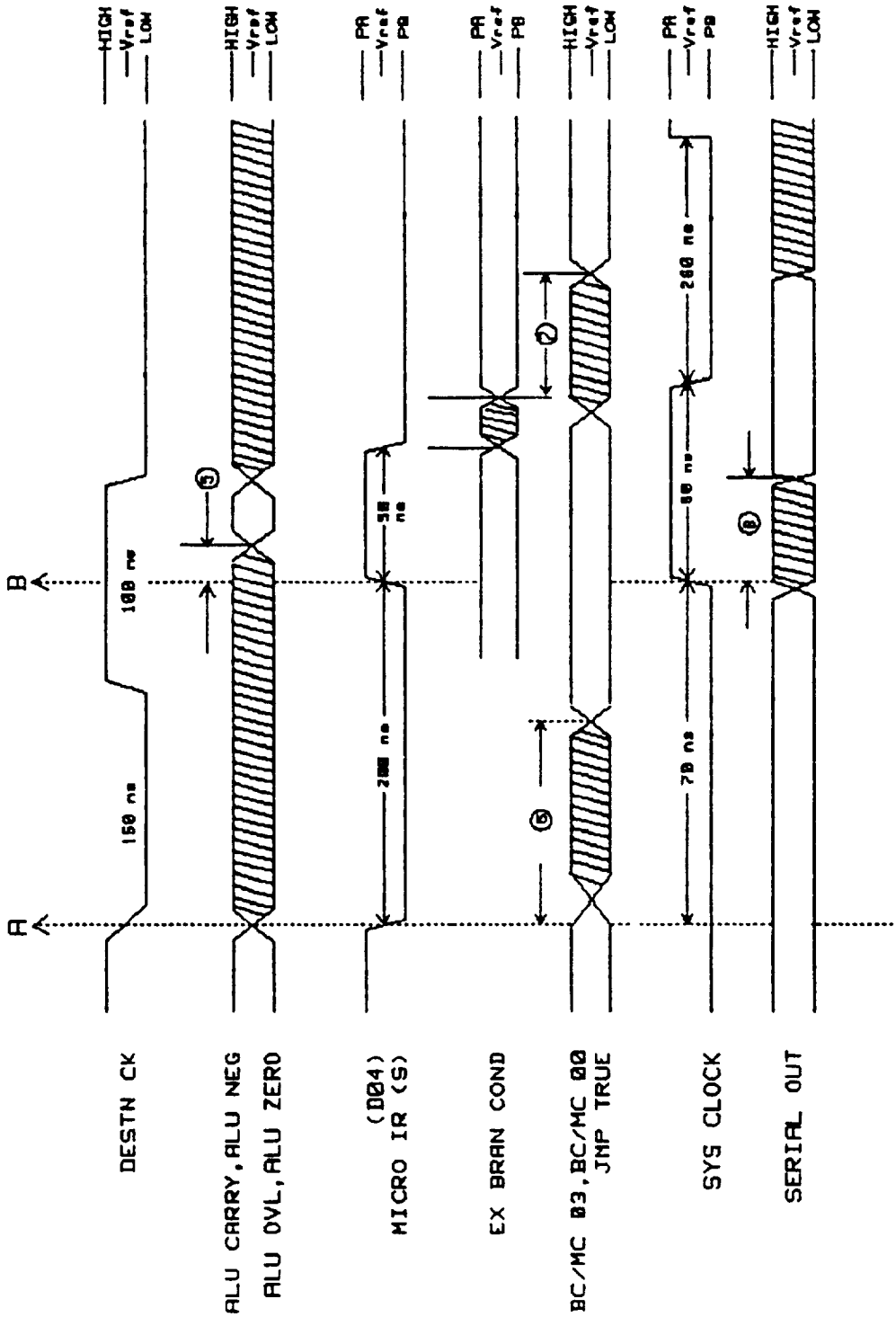


FIGURE 12b. Dynamic waveforms for table XVII.

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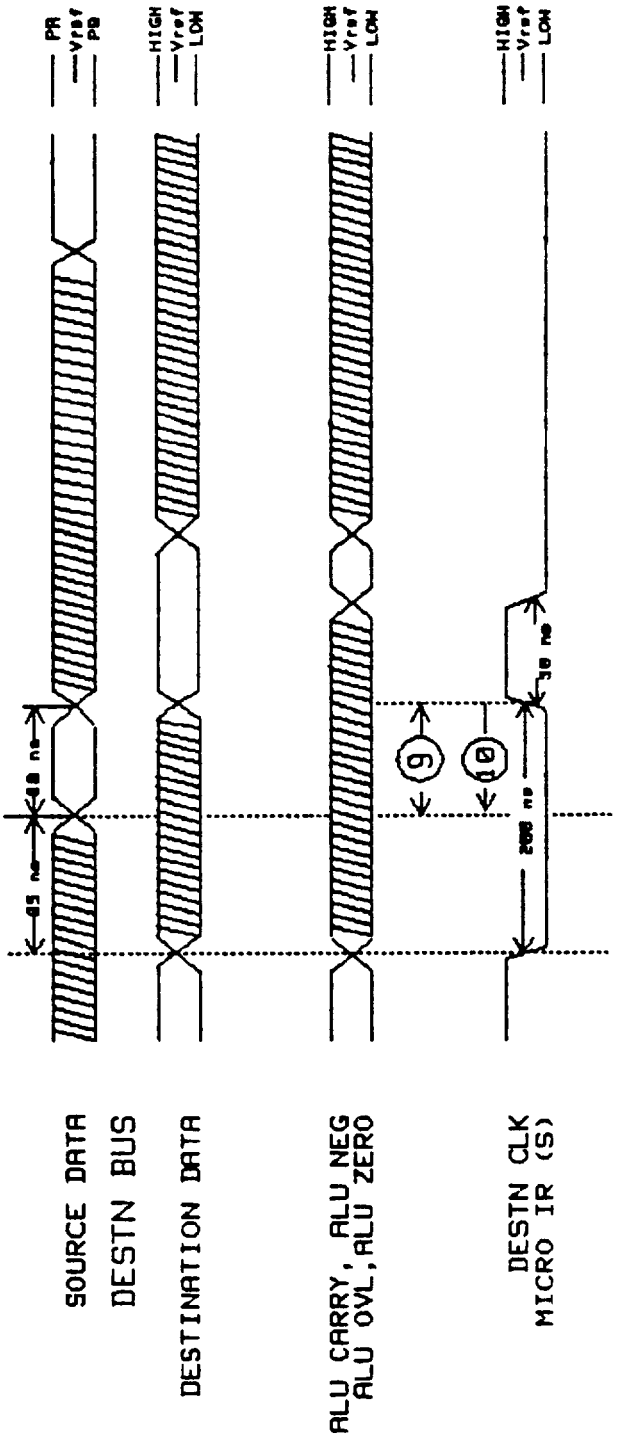


FIGURE 12c. Dynamic waveforms for table XVII.

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TABLE XVIII. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
Inputs						
R1	SOURCE A 15	2	2	50	pF	1-4
R2	SOURCE A 14	3				
R3	SOURCE A 13	4				
R4	SOURCE A 12	5	2	50	pF	1-4
R5	BC/MC 03	19	4	100	pF	1-4
R6	BC/MC 02	20				
R7	BC/MC 01	21				
R8	BC/MC 00	22	4	100	pF	1-4
R9	EX BRAN COND	23	2	75	pF	1-4
R10	MASTER CLEAR	24	2	50	pF	1-4
R11	(SB04)	26	4	75	pF	1-4
R12	MC CL FORMAT	27	2	50	pF	1-4
R13	MC IR (F00)	28	2	50	pF	1-4
R14	TOGGLE ALU	29	4	75	pF	1-4
R15	MCU (00)	30	2	75	pF	1-4
R16	MCU (01)	32	2	50	pF	1-4
R17	(D00)	33	4	75	pF	1-4
R18	(SB03)	35	4	100	pF	1-4
R19	(SB02)	36	4	75	pF	1-4
R20	(SB01)	37	4	75	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R21	(SB00)	38	4	75	pF	1-4
R22	(D01)	39	4	75	pF	1-4
R23	(F03)	41	6	200	pF	1-4
R24	(F02)	42	10	150	pF	1-4
R25	(F01)	43	6	175	pF	1-4
R26	(SA02)	44	4	150	pF	1-4
R27	(SA01)	45	8	150	pF	1-4
R28	(SA00)	46	4	150	pF	1-4
R29	SOURCE A 11	47	2	50	pF	1-4
R30	SOURCE A 10	48				
R31	SOURCE A 9	49				
R32	SOURCE A 8	50				
R33	SOURCE A 3	52				
R34	SOURCE A 2	53				
R35	SOURCE A 1	54				
R36	SOURCE A 0	55				
R37	CLK RATE 2	56				
R38	CLK RATE 1	57				
R39	CLK RATE 0	58				
R40	20 MHZ OSC	63	2	50	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R41	DESTN CLOCK	64	4	100	pF	1-4
R42	(SA03)	66	6	150	pF	1-4
R43	B BUS ENABLE	67	2	100	pF	1-4
R44	WAIT CLOCK	69	2	50	pF	1-4
R45	WAIT REQUEST	71				
R46	READY	72	2	50	pF	1-4
R47	SMI DATA OUT	76	2	75	pF	1-4
R48	MICRO IR (S)	77	2	75	pF	1-4
R49	SMI LD/SF CT	78	4	100	pF	1-4
R50	DISABLE ALU	79	2	75	pF	1-4
R51	(D02)	80	4	75	pF	1-4
R52	(D03)	82	4	75	pF	1-4
R53	(D04)	83	4	100	pF	1-4
R54	ENABLE CHAIN	84	8	100	pF	1-4
R55	(C0)	85	4	75	pF	1-4
R56	(C1)	86	6	100	pF	1-4
R57	SYS CLOCK	92	2	75	pF	1-4
R58	SOURCE A 7	97	2	50	pF	1-4
R59	SOURCE A 6	98				
R60	SOURCE A 5	99	2	50	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R61	SOURCE A 4	100	2	50	pF	1-4
Bidirectionals						
R62	DESTN BUS 15	9	10	150	pF	1-4
R63	DESTN BUS 14	11	8	150	pF	1-4
R64	DESTN BUS 13	12	8	125	pF	1-4
R65	DESTN BUS 12	13	8	125	pF	1-4
R66	DESTN BUS 11	15	8	150	pF	1-4
R67	DESTN BUS 10	16				
R68	DESTN BUS 9	17				
R69	DESTN BUS 8	18				
R70	DESTN BUS 7	87				
R71	DESTN BUS 6	88				
R72	DESTN BUS 5	89				
R73	DESTN BUS 4	91				
R74	DESTN BUS 3	93				
R75	DESTN BUS 2	94				
R76	DESTN BUS 1	95				
R77	DESTN BUS 0	96	8	150	pF	1-4

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NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R77
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R77
Capacitance measurement	NA	± 2.4 pF	R1-R4
	NA	± 4.8 pF	R5-R8
	NA	± 3.65 pF	R9
	NA	± 2.4 pF	R10
	NA	± 3.55 pF	R11
	NA	± 2.4 pF	R12,13
	NA	± 3.55 pF	R14
	NA	± 3.65 pF	R15
	NA	± 2.4 pF	R16
	NA	± 3.55 pF	R17
	NA	± 4.8 pF	R18
	NA	± 3.55 pF	R19-R22
	NA	± 9.7 pF	R23
	NA	± 7.0 pF	R24
	NA	± 8.45 pF	R25
	NA	± 7.3 pF	R26
	NA	± 7.1 pF	R27
	NA	± 7.3 pF	R28
	NA	± 2.4 pF	R29-R40
	NA	± 4.8 pF	R41
	NA	± 7.2 pF	R42
	NA	± 4.9 pF	R43
	NA	± 2.4 pF	R44-R46
	NA	± 3.65 pF	R47,R48
	NA	± 4.8 pF	R49
	NA	± 3.65 pF	R50
	NA	± 3.55 pF	R51,R52
	NA	± 4.8 pF	R53
	NA	± 4.6 pF	R54
	NA	± 3.55 pF	R55

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Capacitance measurement	NA	± 4.7 pF	R56
	NA	± 3.65 pF	R57
	NA	± 2.4 pF	R58-R61
	NA	± 7.0 pF	R62
	NA	± 7.1 pF	R63
	NA	± 5.85 pF	R64,R65
	NA	± 7.1 pF	R66-R77

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. The test circuit shall be as indicated on figure 13.

4. Abbreviations used are defined in 6.1.

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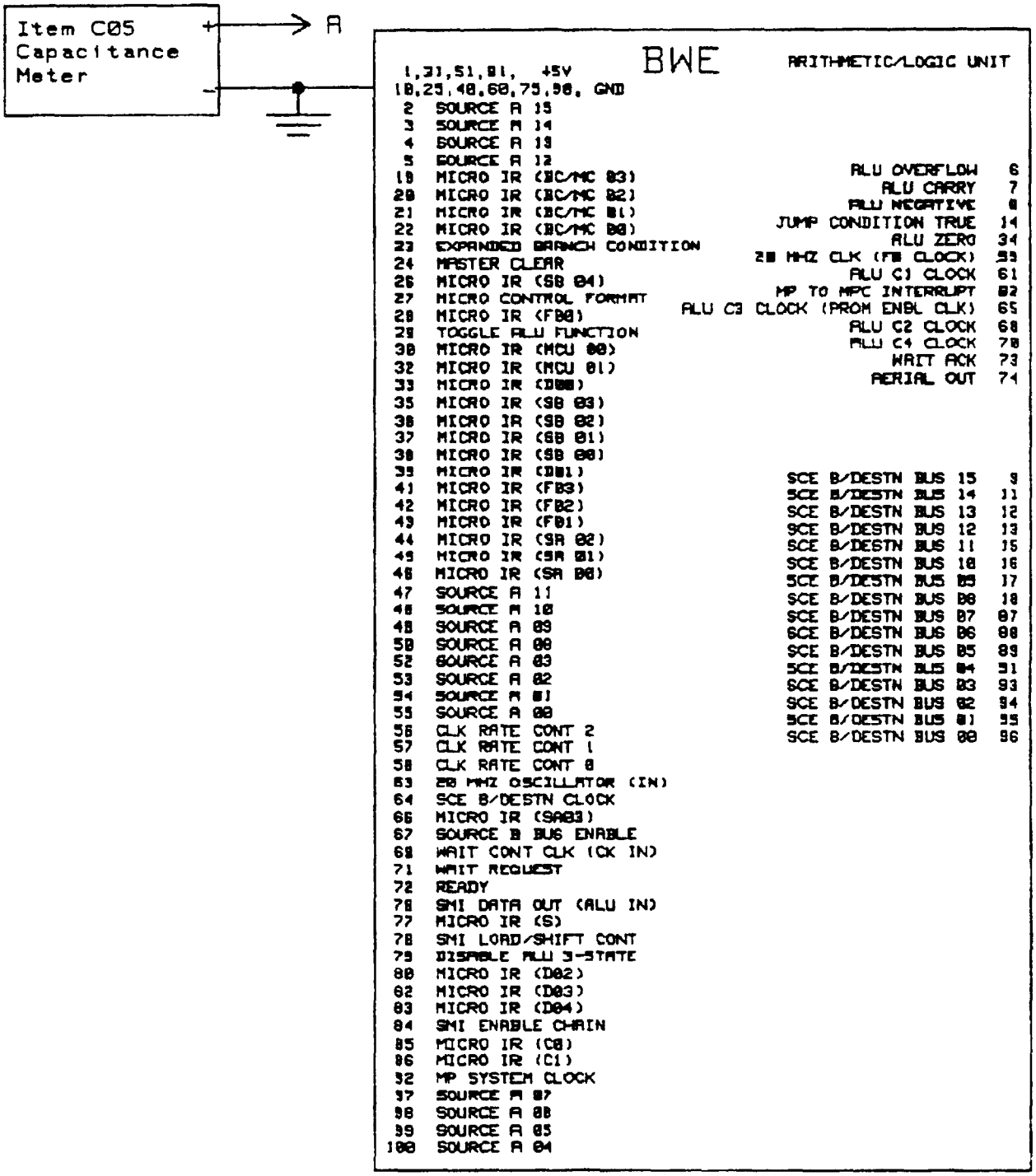


FIGURE 13. Input and output capacitance test circuit for table XVIII.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
T1	PSTA	1	0	400	mV	1-4	14,15
T2	PSTA	51	0	400	mV	1-4	14,15

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T1,T2
Power supply current limit	2.7 A	NA	T1,T2
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T1,T2 T1,T2
Power supply transient amplitude measurement	NA	± 20 mV	T1,T2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX.

(c) Vcc pins 1 and 31 shall be shorted together, Vcc pins 51 and 81 shall be shorted together and the scope point, probe 'A', connected to the pin indicated in table XIX.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 14 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 15.

4. Abbreviations used are defined in 6.1.

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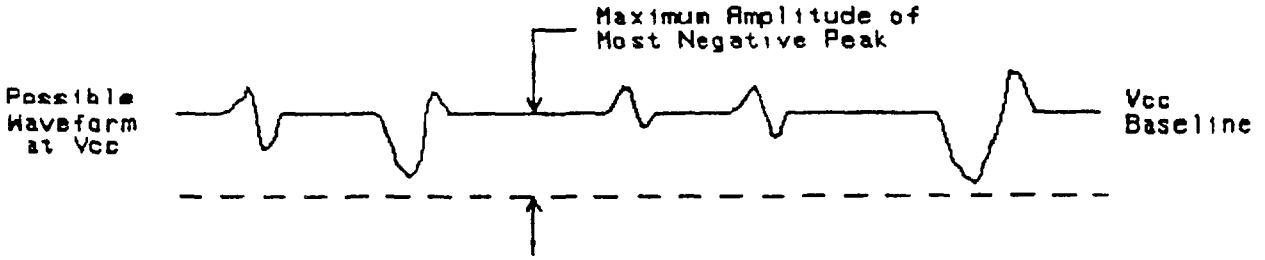
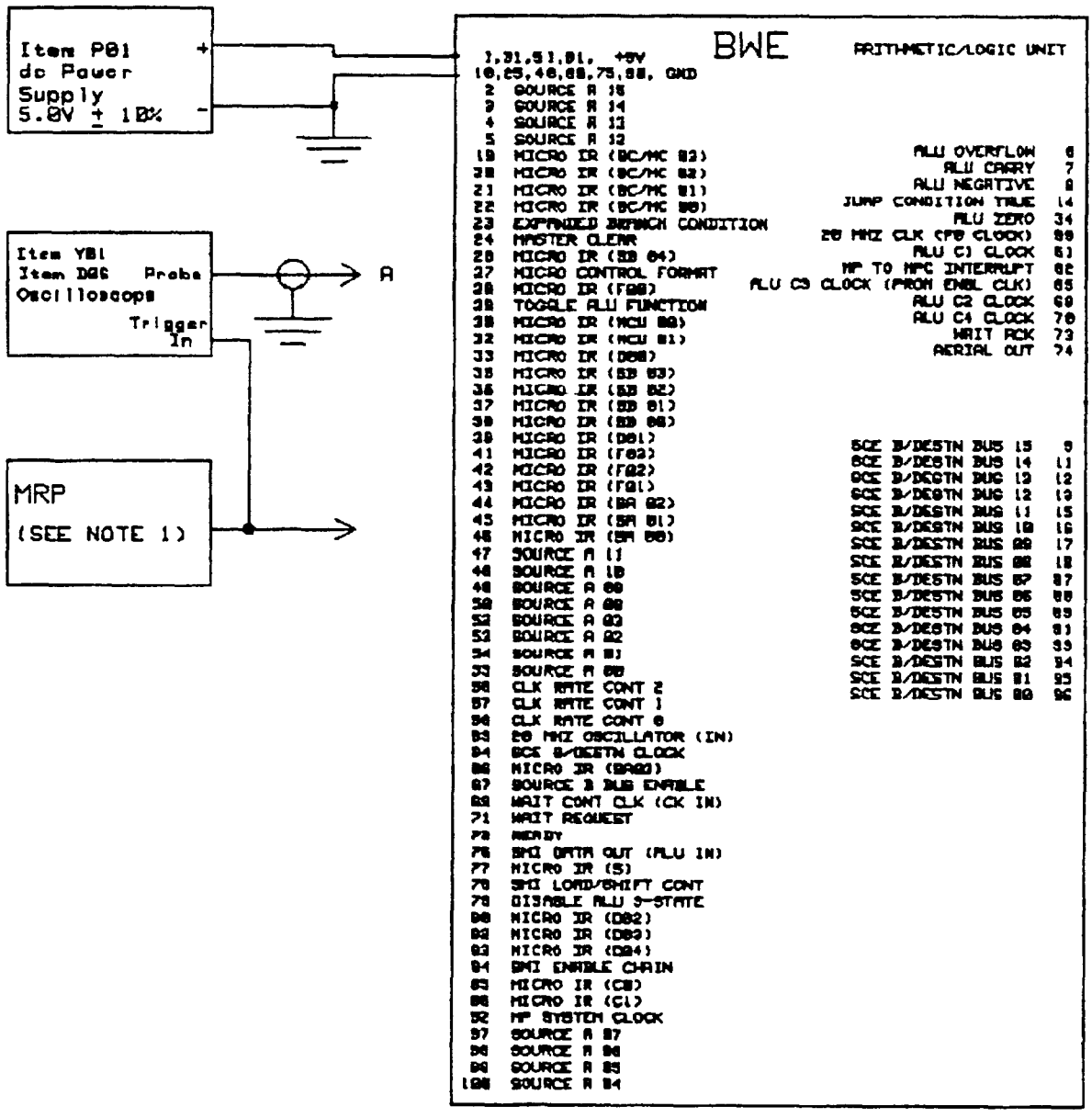


FIGURE 14. Power supply transient amplitude waveform for table XIX.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 15. Power supply transient amplitude test circuit for table XIX.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-28787/510	Modules, Standard Electronic Arithmetic-Logic Unit Module, Key Code BWE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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